המכון למחקרי נפט וגיאופיסיקה
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

Israel

Geothermal Project

Electrical Resistivity Survey

The Yizreel and Harod Valleys

by

Y. Rotstein, A. Klang and A. Levanon

Prepared for

The Office of Chief Scientist
Ministry of Industry and Commerce

March, 1977

GT/149/74 (2)

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ABSTRACT

As part of the geothermal program in Israel, an electrical resistivity survey was conducted in the Yizreel and Harod valleys and in nearby areas. The survey, which was designed as a reconaissance survey, used the roving dipole, dipole-dipole and sounding methods. Roving dipole sources were designed so that most areas were covered by different sources and looked upon from different directions, so as to minimize the effect of localized conditions.

As a result of this survey, many relatively lower apparent resistivity areas were defined and described. Most, however, were observed using only one source, and were not confirmed by additional data. Only a few areas stood out as areas of possible low resistivity, and these could also be explained without the presence of geothermal reservoirs.

The results of this survey show that on the whole, the Yizreel and Harod valleys should not be emphasized in future geothermal exploration. Some limited additional work is recommended, however. The work should include the drilling of two shallow temperature holes near Yokne'am, and possibly two additional ones near Gid'ona.

1.0 INTRODUCTION

As part of a project destinated to find geothermal resources in Israel, the Institute for Petroleum Research and Geophysics (I.P.R.G.) conducted electrical reconnaisance surveys. These are to be followed by additional surveys, including detailed electric surveys, in areas which prove to be of geothermal interest.

This report will describe the results of an electrical survey of the Yizreel and Harod valleys (Figure 1.) The surveyed area, a recent graben, is considered a potential area for geothermal resources due to the young volcanism which developed concurrently with the faulting. This volcanism appears as extended basalt flows known as the Lower and Cover Basalt of Neogene-Pleistocene age. Also present are igneous intrusive bodies as dykes, stocks and pipes which penetrated into the sediments (Dicker, 1969 and Figure 1 of this report.)

Another factor of interest is the presence of the Shoqeq Spring (Figure 1) which has a temperature of about 29°C, as compared with the average temperature of 22°C usually found in the valley.

The survey was conducted during the periods of July-November, 1975 and August - November, 1976. The electrical methods used in this survey were the roving dipole mapping, dipole-dipole profiling, modified Schlumberger and monopole soundings. Seventeen "L" configuration dipole sources were occupied for the roving dipole mapping. A total of 796 receiver stations were measured. Five dipole-dipole profiles and sixty eight modified Schlumberger and monopole soundings were carried out. The dipole-dipole profiling was conducted with a spacing of 500 m. and with N of up to 8.

This report will first describe the known geological and geophysical data available from this region. Next, a brief description of the methods will be presented, followed by a detailed discussion of the results, their interpretation and the conclusions which may be drawn from them.

2.0 GENERAL GEOLOGICAL BACKGROUND

The Yizreel and Harod valleys are part of a graben which was probably formed at the end of the Tertiary and the beginning of the Quaternary by intense fracturing and graben-faulting. The fracturing and faulting took place in all of Northern Israel, and is possibly connected with the subsidence of the Jordan-Dead Sea graben (Kashai, 1974).

In the south and southwest, the graben is bordered by the Carmel horst and more specifically by the Umm el Fahm anticline. To the north, it is bordered by a fault system and the Debora structure.

Most of the area is covered by recent and sub-recent alluvial deposits and soil, with a few surface exposures of older rocks such as Eocene, Senonian and Upper Cenomanian (Figure 1.). Basaltic flows cover the eastern part of the surveyed area, and similar isolated basaltic appearances are found west of Afula.

The geology of the area can be estimated with the help of the detailed geological mapping around the Emeq, and by the data from deep wells (Devorah 1 and Sarid 1), stratigraphic wells (Gid'on 1-5) and water wells.

A detailed discussion of the Emeq geology is given by Eliesery (1964) and Dicker (1969). The general lithology of the various rock units is as follows:

Jurassic: limestone, basalt and tuffs interbedded with limestone
Lower Cretaceous: interbedding of shale and some limestone
Upper Cretaceous:
Cenomanian: mainly dolomites, chalk and limestone with chert nodulus, possibly tuff and basalt.
Turonian: limestone.
Senonian: mainly chalk, occasionally bituminous, chert and marls.

Paleogene: limestone, chalk.

Neogene: basalts (Lower and Cover Zones) with a clay series between the two zones.

The Lower Basalt zone has a variable thickness (200 to over 500 m.) and is composed of alkali olivine basalt.

The Upper (Cover) Basalt is composed of hard gray olivine basalt and is exposed in the eastern part of the area (north of the Afula-Bet She'an road). Its northern boundary is Nahal Yosef, where it lies directly on the Lower Basalt. It is found also in drillholes near Afula (less than 20 m. thickness.)

In addition to the Basalt flows, other igneous intrusive bodies are exposed in the Emeq (Givat Hamoreh and Indur.) Dykes intrude into Eocene Limestone sequence and can be grouped in two systems, one oriented ENE and the other WNW. Stocks and pipes are also present.

The intrusions are closely related to the distinct structural elements, but there is no sign of any deformation brought about by the intrusion process itself (Dicker, 1969.)

The dominant structural elements in the area are the faults (Post Middle Eocene to Upper Tertiary and Quaternary). The resulting structures are tilted blocks (Qara, Qumi), horsts (Givat Ha'moreh, Indur), folds central Emeq anticline of the Umm el Fahm - Debora structural line), and depressions (Afula).

Numerous springs are found in the region, located mainly on the northern and south--southwestern edges of the graben and probably lying along the fault system limiting the depression. Southwest of Afula the springs extend into the graben. The salinity of the spring water is variable, but in general is less than 290 mg Cl/l with the exception of the springs located near Bet Alfa which have a salinity range of 600-1060 mg Cl/l

(Rosenthal, 1965, Dilinski and Kotin, 1970). In general, the water attains a temperature of about 20-22°C, with the exception of the En Shoqeq spring, (Figure 1) which reaches a temperature of about 29°C (Rosenthal, personal communication).

3.0 PREVIOUS GEOPHYSICAL SURVEYS

3.1 Magnetometry An aeromagnetic survey covering Israel was interpreted by Domzalski (1967). The survey was completed in 1970, to cover all of the Judea and Shomron mountains. The integrated data was interpreted by Folkman (1971).

In general, the magnetic intensity map shows the effects of both shallow and deep causative bodies.

The known volcanic outcrops in the Emeq Yizreel area correspond to pronounced anomalous zones. These zones, however, extend over a much larger area than shown by the outcrops, indicating the presence of near surface volcanics under much of the valley. Domzalski (1967) has indicated the possible existence of fractures in the area, which provide openings for the volcanics. He concludes that the volcanics which are present are of "local" origin, rather than flowing from distant sources.

Other trends in the magnetic map, defined by contour flexures, are less definite and may signify subsidiary fractures and deeper seated intrusive bodies of linear character.

Folkman (1971) remarks that the western boundary of the graben is not detected by the map, probably because of its gradual slope which is obscured by the over-riding magnetic effects of shallow origin.

A positive anomaly exists around the town of Afula.

3.2 Gravimetry A gravity survey of the region is described by Ginzburg (1960, 1969). According to his interpretation, the western part of the Depression is covered by a large circular low centered near Ramat David. This low is bordered to the west by the gradient of the Carmel which is associated with the down faulting towards the graben.

An elongated gravity high starting from Megiddo and extending to about 6 km. northwest of Afula, appears in the central part of the Emeq. This anomaly represents the continuation of the Umm el Fahm structure and was interpreted by Ginzburg (1960) as having the following explanations:

1. The gravity high is caused by dense Turonian-Cenomanian uplift.
2. A thick basalt layer exists directly beneath the crest of the anomaly, but rapidly thins near the edges.
3. A combination of 1 and 2. This possibility is considered the most probable.

North of this gravity high, a gravity low directed east-west and extending up to the strong gradient which appears over the edge of the graben, represents the extension of the western depression to the Tabor Depression. The thickness of the post-Turonian deposition under the center of the anomaly was estimated by Ginzburg (1960) to be about 500 m.

A few local highs are also present north and northeast of the main high, such as the one south of the Muallaga anticline and another south of Devora.

A large low appears around Afula and extends southeast towards Bet She'an.

A prominent gravity high, produced by the rise of the Gilboa, with the general direction of southeast-northwest, is located at the flank of the mountain. This high also extends in the southwest direction towards Jenin.

A low appears south of Afula on the western side of the mentioned high, extending to the Jenin area as well.

3.3 Seismics The available seismic data obtained in a survey conducted in 1970, is interpreted by Klang and Sherman (1972). Only three lines were surveyed, showing strong reflectors which lack good continuity. Due to these facts, no map could be constructed.

The general picture obtained by this survey shows a large positive structure, bordered to the south-southeast by a fault passing through the Sarid 1 well. The fault appears to change strike from southwest to east-northeast in the vicinity of the Sarid 1 well. About 1 km. southeast of this fault another fault with a similar trend is interpreted. An indication of positive structure is obtained on a seismic line northeast of Afula also. This seismically interpreted structure appears to correspond with the gravity high (Megiddo-North Afula).

Experiments to improve the quality of seismic data were conducted by the I.P.R.G. during 1972 using detonating cord, and again in 1975 employing the "Vibroseis" system. However, no improvement was achieved in these surveys.

In 1976, Oil Exploration (Investments) conducted a seismic survey in the area, using the "Vibroseis" system. The data is currently in the phase of being processed and interpreted.

3.4 Magnetotellurics The available data from a limited magnetotelluric survey conducted in the Emeq Yizreel area are interpreted by Klang and Sherman (1972).

The results indicated relatively low resistivity values (4-10 ohm-m.) near surface which could be correlated with the Neogene clays, shales and sandstones found in the Sarid 1 and Gid'on 5 wells. Higher resistivity values (12-40 ohm-m.), found west of Sarid, could be correlated with the basalt and limestone section found in the Gid'on 1 well.

In the area between the Sarid 1 and Gid'on 5 wells, this lower resistivity layer overlies a high resistivity layer of 25-50 ohm-m. which corresponds to the limestone section of the Lower Cretaceous and Jurassic, as found in the Sarid 1 well, and to the basalt in Gid'on 5. Below this layer, at a depth of about 1750 to 2200 m., a low resistivity layer (4-7 ohm-m.) may exist. This is again followed, at a depth of about 3000 to 4500 m., by a high resistivity layer (112-126 ohm-m.).

A possible fault was interpreted as the continuation of a fault encountered at the Sarid 1 well, which was also found by the seismic survey. Some uncertainty exists as to the accuracy of this detailed resistivity section due to problems associated with quality of the field data.

3.5 Electrical Resistivity Electrical resistivity surveys using the Wenner and Schlumberger sounding methods were conducted in the Emeq Yizreel area by the I.P.R.G. during the period of 1961-1976. A large number of soundings were measured with short spreads (about 200-400 m.) and only a limited number of soundings used longer spreads and extended to greater depths ($\frac{AB}{2}$ about 800-1000 m.).

In the central Emeq area (Gid'on wells), the general resistivity section found by the deep soundings (Levanon, 1964) could be described as follows:

a. A thin (about 10-50 m.) layer with resistivities of 3 to 9 ohm-m., corresponding to the soil and alluvial coverage.

b. A second layer with resistivities in the range of 30 to 90 ohm-m. and with a thickness of about 250 to 400 m., corresponding to basalt flows.

c. A third layer with high resistivities (160-250 ohm-m.) and with unknown thickness which corresponds to a limestone or dolomite sequence.

In the Megiddo area the general resistivity section found by some Wenner soundings (a = 500 m.) is as follows (Paparo, 1974);

a. A first layer with resistivities in the range of 15 to 29 ohm-m. and with a thickness of about 100-250 meters.

b. A second layer with resistivities of 40-150 ohm-m. and with undetermined thickness.

In the Emeq Harod area, the resistivity section found by two deep soundings could be divided as follows (Levanon, 1976);

a. A thin (about 9 m. thick) layer with resistivity of about 4 ohm-m. corresponding to the soil and alluvial deposits.

b. A second layer with resistivities of 140 ohm-m. and with a thickness of 80 meters, corresponds to the Eocene limestone.

c. A third layer with resistivities of 20 ohm-m. and with a thickness of 360 m., corresponds to the Senonian chalk.

d. A fourth layer with resistivities of about 100 ohm-m. corresponds to the Upper Cretaceous limestone, but its thickness has not been determined.

The resistivity section found by the second sounding is as follows:

a. A first layer with resistivities of 10 to 20 ohm-m. and with a thickness of about 60 m.

b. A second layer with a resistivity of 60 ohm-m. and with a thickness of 440 m. corresponding to a chalky limestone section.

c. A third layer with a resistivity of 370 ohm-m. corresponding to limestone and dolomite, and with undetermined thickness.

The relationship between the electrical resistivity values and the basalt aquifers in Israel, including Emeq Yizreel, were studied by Levanon

and Barner (1970). They concluded that the calculated resistivity values for the lower Basalt aquifer vary from 60 to 165 ohm-m., with values of 20 to 30 ohm-m. for dry wells, and that the calculated resistivity range from 15 to 70 ohm-m. for the Upper Basalt.

The higher resistivity values of the Lower Basalt, as compared to those of the Upper Basalt, may possibly be explained by some difference in mineralogical composition or by a process of secondary mineralization of the older baslt.

There is an interdependence between the specific yield and the actual resistivity values in the basalt aquifers. The resistivity values are being influenced by the porosity and lithology of the basalt. At the same time, salinity of up to 5000 mg/l TDS has been found to have no effect upon the resistivity values of the basalt aquifers.

4.0 ELECTRICAL METHODS USED

The electrical methods used in this survey were roving dipole mapping, dipole-dipole profiling and the modified Schlumberger and monopole soundings.

4.1 Roving Dipole The roving dipole (Figure 2) is designed for rapid reconnaissance of large areas in order to select specific targets for detailed investigation.

In this method, a long (about 2-3 km.) source dipole (AB) is used with a high current density to produce a very large electric field. Potential dipole (NMN) has an "L" shape so as to produce a resultant vector resistivity. Potential dipole sites are distributed around the source at various asimuths and distances, to produce an apparent resistivity map. In contrast to other methods for mapping lateral resistivity variations, with the roving dipole mapping method, the source electrodes remain fixed in location and only the receiver electrodes are moved. Hence, the variations in electric field behavior between adjacent receiver stations can be identified

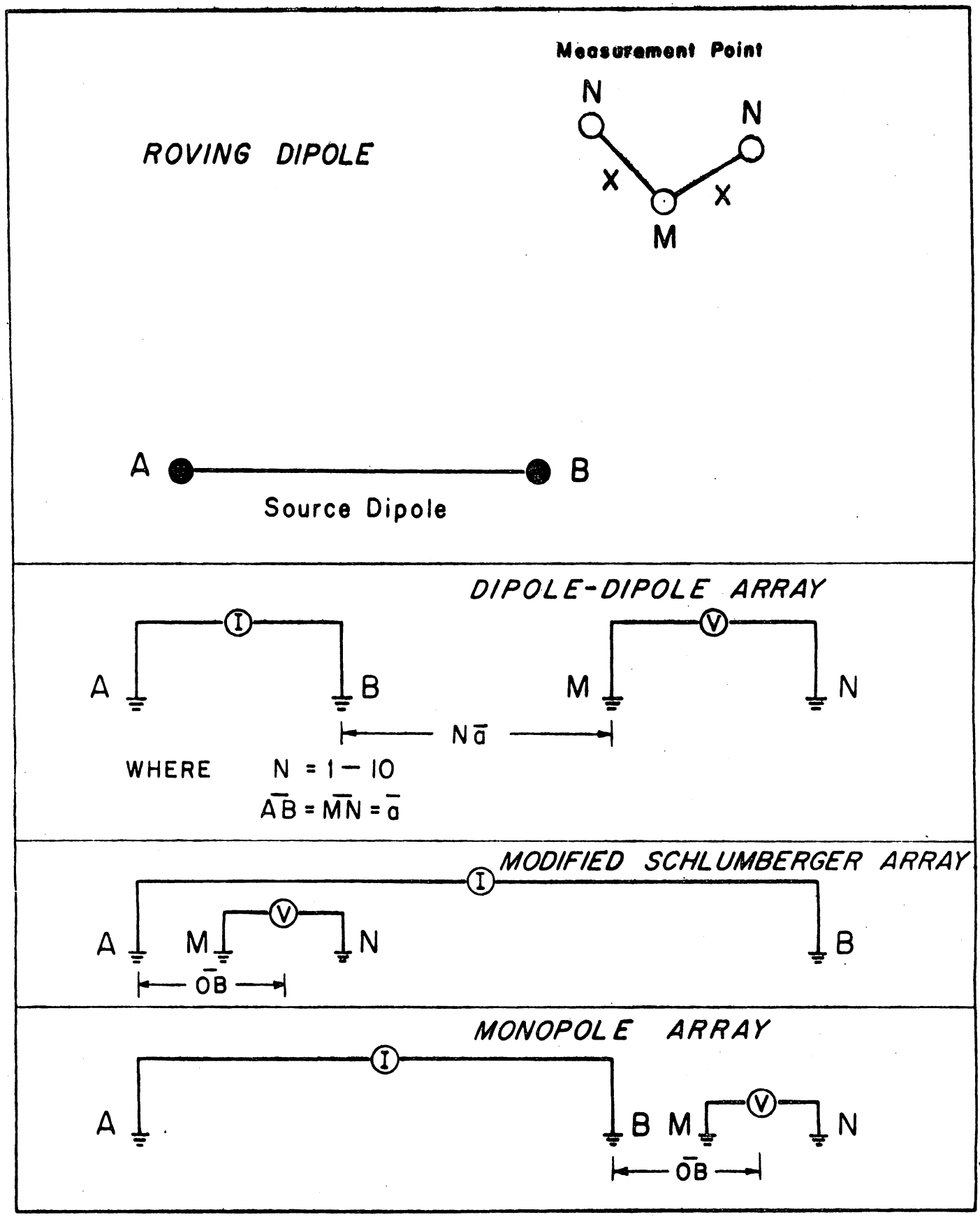


Figure 2

TYPES OF ELECTRIC RESISTIVITY CONFIGURATION USED

with changes in earth resistivity immediately in the neighborhood of the receiver stations, rather than being a function of earth properties at the source location or at points between the source and receiver. The method is often used in geothermal exploration and it is described by many authors, including Risk et al. (1970) and Keller et al (1975). This method is generally used for defining the outline of large and very conductive bodies.

Low resistivity anomalies mapped by the roving dipole method can be the expression of lateral resistivity changes due to variations in rock types and characteristics, but they can also be the result of water present in the aquifer. The increased salinity may be caused by significant increase in subsurface temperatures.

In using the roving dipole method, one applies a multiple illumination of the area. This illumination is obtained by repeated occupation of the receiver stations for source dipole in different places and by the usage of "L" source type configuration in each place. Multiple illumination may provide one or more of the following:

a. Separations of surface effects from effects of deeper rocks.

b. Estimation of depth to electrical basement.

c. Location of major boundaries and faults.

d. Estimation of resistivity contrast between resistive and conductive areas.

4.2 Dipole-Dipole The dipole-dipole profile (Fig. 2) had long been used in mineral exploration as an effective detailing method. In this method, both the source dipole (AB) and the potential dipole (MN) are moved along the profile. Increased depth information is obtained by moving the potential dipole at increasing distances from the source. The results are presented in the form of an apparent resistivity cross-section. The vertical

coordinate of this cross-section is not expressed in terms of depth, but rather as a function of increased distance between dipoles. A discussion of the dipole-dipole method is presented by Keller (1966).

4.3 Depth Soundigs A variation of the Schlumberger configuration is the modified Schlumberger configuration (Fig. 2). The potential dipole (MN) is small as compared to the source dipole (AB) and is expanded co-linearly from one source electrode towards the center of the source dipole. The approximate depth of penetration in homogeneous earth is usually taken to be equal to the distance OB, whenever $OB << \frac{AB}{2}$.

In the monople configuration (Fig. 2) the source dipole length (AB) is assumed to be very great (infinite) when compared to the potential dipole length (MN) and to the distance OB. The potential dipole is expanded away from the source dipole's center (respectively from the source electrode) along an asimuth equal to that of the source dipole. The effective probing depth (as defined by Keller, 1966) of this configuration in homogenous earth is equivalent to the separation OB.

5.0 RESULTS

5.1 Roving Dipole Maps Seventeen dipole sources ("L" type configuration) were used in the mapping of the apparent resistivity in the Emeq Yizreel and Emeq Harod areas. The location of all the sources is shown in Fig. 1. The apparent resistivity maps obtained for the various sources are described individually. In particular areas of lower apparent resistivities relative to their surroundings will be described. Their possible geothermal implications will be discussed later in this report.

Source 41

The general trend of the resistivity contours for this source indicates an increase in apparent resistivity values from the source in all measured directions (Fig. 3).

A low apparent resistivity zone (A), with values of less than 3 ohm-m. appears near the southern pole of the source. From this zone, further to the south, the resistivity increases with a strong gradient.

Source 42

One low resistivity area (Zone A in Fig. 4) with apparent resistivities lower than 5 ohm-m. appears west of the source, followed by an increase in resistivities further to the west. Another low apparent resistivity area (Zone B) is present east-northeast of the source, and near it. This low is bordered by an increase of the resistivities to the east. At the eastern edge of the map, a decrease in resistivities is apparent. Two small low apparent resistivity areas, C and D are defined. In the southerly direction, a general increase in apparent resistivity values is observed.

Source 43

In the apparent resistivity map of this source, a low resistivity zone located south of the source (Zone A in Fig. 5) is present with apparent resistivity values of less than 7 ohm-m. It is possible that these low resistivity values extend further to the south.

In the easterly direction, the resistivity values incease with increased distance from the source. West of the source high values of resistivity are apparent in the source's immediate vicinity. Further to the west resistivity decreases, but the extent of the decrease is not clearly defined.

Source 44

Observed resistivity values are in general over 20 ohm-m. and do not show a specific pattern. A relatively low resistivity closure is apparent

about 2 km. south of the source (Zone A in Fig. 6), but the observed values of over 10 ohm-m. are not very low. The source itself is located in a high apparent resistivity area.

Source 45

A limited area, mostly to the southeast, was surveyed with this source. An area with apparent resistivities lower than 15 ohm-m. (Zone A on Figure 7) appears south of the source. From this area, the resistivity values increase westwards, and in the northeasterly direction as well.

A one point low apparent resistivity zone (Zone B in Fig. 7) is found at the southern edge of the map. It is possible that this area extends further south beyond the limits of the mapped area.

Source 46

The survey around this source was done mostly to the southeast. A large low apparent resistivity closure (Zone A in Fig. 8) with values of less than 10 ohm-m. appears to the southeast of this source. The extension of this zone to the south and west is not clear.

To the east and north of this area, resistivity values rapidly increase.

Source 47

An elongated low resistivity area (Zone A in Fig. 9), with resistivity values of less than 7 ohm-m., is mapped near the source. From this area, the resistivity values increase in all directions, reaching values of over 50 ohm-m. at the eastern edge of the map.

Source 48

A large low resistivity area (Zone A on Fig. 10) with apparent resistivity values of less than 10 ohm-m. extends from the source northwards,

to the edge of the map. From this area, the resistivity values increase in all directions with a more pronounced gradient to the west.

Source 49

A one point low apparent resistivity value of less than 10 ohm-m. possibly indicates the presence of a low resistivity area (A in Fig. 11) near the source. This area could not be clearly defined due to the lack of measurements in the vicinity of the source.

Apparent resistivity values increase with increased distance from the source in all directions, with a more pronounced gradient in the south-southeast direction.

A local decrease in resistivity values is observed at the north-western edge of the map. A one point low resistivity value of less than 7 ohm-m. indicates the possible presence of a low resistivity area (Zone B in Fig. 11) apparently extending further northwards from the mapped area.

Source 50

The map, which covers some 150 square km. south of Afula, differs from the preceeding one in that it includes a number of low resistivity areas.

An apparent low resistivity area with resistivity values of less than 15 ohm-m. (Zone A in Fig. 12) appears near the source and extends to the north. This apparent resistivity low is not well defined in the areal extent due to the lack of measurements near the source.

The resistivity values increase south and west of the source. Towards the east, the apparent resistivity increase is less pronounced and 3 local low apparent resistivity areas with resistivity values of less than 15 ohm-m. (Zones B,C, and D in Fig. 12) are also mapped.

A local decrease in the resistivity values at the northeastern edge of the map suggests the presence of another low apparent resistivity area with values of less than 15 ohm-m. (Zone E in Fig. 12), possibly extending further northwards outside the existent map.

Source 51

High resistivity values (over 25 ohm-m.) appear in the entire area mapped by this source (Fig. 13). A general increase of the resistivity values from the source in the westerly and easterly directions is indicated by the apparent resistivity map. From about 30 ohm-m. around the source, the resistivity values reach about 130 ohm-m. at the western edge of the mapped area and about 70 to 100 ohm-m. at the eastern borderline of the mapped area.

Source 52

The general resistivity distribution for this map (Fig. 14) is the same as that described for the map of source 51. The apparent resistivity values increase from the relatively high value of 40 ohm-m. near the source, to about 100 ohm-m. in the west and east, at the borders of the mapped area.

Source 53

A relatively low apparent resistivity area (Zone A in Fig. 15) with resistivity values of less than 15 ohm-m. appears at the southern pole of the dipole source. From this area, further to the south, the apparent resistivity values increase to a value of 114 ohm-m. at the southern edge of the mapped area. An increase of apparent resistivities is observed also west and east of the source; the highest resistivity values of about 170-190 ohm-m. are situated at the eastern edge of the mapped area.

Source 54

In general, apparent resistivities mapped by this source are significantly lower than those mapped by source 53. Two low apparent resistivity areas (Zone A and B in Fig. 16) with resistivity values of less than 7 ohm-m. are located west of the source. Further to the west and southwest the resistivity values increase.

Another relatively low apparent resistivity area, (Zone C in Figure 16) with values less than 10 ohm-m. appears near the eastern pole of the dipole source and extends eastwards, possibly beyond the mapped area. North and south of this area, the resistivity values increase.

Source 57

Apparent resistivity values around source 57 are mostly between 30 and 70 ohm-m. with a tendency towards increased values in the vicinity of the source (Figure 17). About 2 km. to the south of the source, a relatively low apparent resistivity area (Zone A in Fig. 17) with resistivity values of less than 15 ohm-m. is present. Further to the south, the resistivity values increase to about 70 ohm-m. at the edge of the mapped area. The resistivity values also increase west of the source. At the southeastern corner of the mapped area, a steep gradient of increasing resistivity values is observed.

Source 58

The source is situated in a high apparent resistivity area, (Fig. 18) and from it apparent resistivity values decrease in all directions. The decrease is particularly evident towards the southeast, where values reach a minimum of less than 10 ohm-m. centered in two zones situated about 3 km. from the source (zones A and B in Fig. 18). In all other directions the resistivity reaches 20-30 ohm-m.

Source 59

About 3 km. north and 2 km. northeast of the source appear two zones (A and B in Fig. 19) with apparent resistivities of about 8 ohm-m. and less than 15 ohm-m., respectively. From these areas the resistivities increase towards the east and west with a steep gradient to the west. Another area with apparent resistivity values of less than 15 ohm-m. appears at the south-western corner of the mapped area. This area is bordered by a strong gradient of increased apparent resistivity to the north and south-west.

Source 60

A low apparent resistivity area (Zone A in Figure 20) with resistivity values of less than 15 ohm-m. appears at the source location. More measurements are needed in order to locate its southerly extension. From this area the apparent resistivity values increase in all directions, but with a more pronounced gradient to the north.

Source 61

Two relatively low apparent resistivity areas (Zones A and B in Fig. 21) with values of less than 10 ohm-m. appear, one near the northern pole of the source and the other about 2 km. further to the northeast. This last zone possibly extends beyond the edge of the surveyed area. From these low areas, the apparent resistivity values increase with a steep gradient to a value of about 100 ohm-m. in the western part of the map.

In the east and southeast, the apparent resistivities increase slowly and only at the southeastern edge of the area do they rapidly increase.

Source 62

A low apparent resistivity area (Zone A in Fig. 22) with resistivity values of less than 10 ohm-m. appears 2 km. northeast of the source. This low extends to the east and appears to continue outside of the mapped

area. Additional measurements are needed in order to find the eastern extension of this low apparent resistivity area. North and south of the mentioned zone, the apparent resistivity values increase to about 30 ohm-m.

The resistivity values also increase westwards of the source, reaching more than 100 ohm-m. at the northwestern edge of the mapped area.

Source 63

The source seems to be located in a high apparent resistivity area (Fig. 23). About 2 km. west of the source, a lower apparent resistivity area (Zone A in Fig. 23) with resistivities of less than 10 ohm-m. is situated. Further to the west, the resistivity values increase up to the edge of the mapped area.

The apparent resistivity values rapidly decrease from about 40 ohm-m. near the source to less than 5 ohm-m. at the southeastern edge of the area; this open zone is marked Zone B in Fig. 23. At the northern edge of the map another low apparent resistivity area is noted (Zone C in Fig. 23).

Source 64

The apparent resistivity map for this source shows a general decrease of the resistivity values from about 30 ohm-m. at the western edge of the mapped area to less than 5 ohm-m. at the eastern edge of the mapped area (Zone A in Fig. 24).

Source 65

Source 65 is located along the border between the Bet She'an and Jordan valleys. The apparent resistivity map for this source shows apparent resistivity to be generally less than 20 ohm-m. (Fig. 25). A general decrease in resistivity values, from the source eastwards and towards the center of Bet She'an and Jordan Valleys, is clearly seen. Two relatively

low resistivity areas (Zone A and B in Fig. 25) with apparent resistivity values of less than 3 ohm-m. appear at the north-eastern and south-eastern edge of the mapped area. The resistivity values also decrease in the southerly direction from the source, and slowly increase in the westerly direction.

Source 66

The apparent resistivity map for this source (Fig. 26) also includes mostly low values. A general decrease in resistivity values from west to east is seen, and at the eastern edge of the map, in the Jordan Valley, apparent resistivity values of less than 2 ohm-m. are observed (Zones A and B in Fig. 26). A localized high exists northwest of the source with apparent resistivity values of over 30 ohm-m.

Source 71

Two areas with apparent resistivity values of less than 10 ohm-m. (Zones A and B in Fig. 27) appear at the southwestern edge of the mapped area. These zones are not well defined due to the lack of measurements.

An increase in the apparent resistivity values is observed eastwards of the source with some local zones of relatively high resistivity values (over 40 ohm-m.). At the eastern edge of the mapped area, the trend of the resistivity values may decrease to form another lower resistivity zone but the information from this area is incomplete.

Source 72

A low apparent resistivity zone (A in Fig. 28) appears near the eastern pole of the source with resistivity values of less than 10 ohm-m. From this area the apparent resistivity values increase in all directions and, in general, apparent resistivity values throughout the map range between 20-40 ohm-m.

Source 73

The apparent resistivity map for this source shows a slow increase in resistivity values to the west and east of the source (Fig. 29). Only at the western and eastern edges of the map a more pronounced gradient is found.

A one point low apparent resistivity zone (A in Fig. 29) with resistivity values of less than 10 ohm-m. appears at the southeastern edge of the map. This zone probably extends beyond the surveyed area, however the lack of measurements in the area prevents its full delineation.

Source 74

The apparent resistivity map for the source indicates an increase in resistivity values from about 20 ohm-m. near the source to about 60 ohm-m. at the eastern border of the mapped area (Fig. 30). A local high is observed at the southwestern corner of the map. No low apparent resistivity was interpreted in this map.

Source 75

An apparent resistivity zone with resistivity values of less than 20 ohm-m. (A in Fig. 31) appears near the source. This zone is not well defined due to the absence of adequate coverage north and west of the source. From this area apparent resistivity values increase rapidly towards the east to a maximum value of about 140 ohm-m. This increase is followed by a decrease in resistivity values to about 40 ohm-m. at the eastern edge of the mapped area.

Source 76

An area with apparent resistivity values reaching less than 20 ohm-m. (A in Fig. 32) appears near the source. However, this zone is mostly defined by one point of less than 20 ohm-m. and the lack of more measurements prevents its full designation.

From the source area the apparent resistivity values increase towards the southeast and northwest. Another one point lower apparent resistivity area with resistivity of less than 20 ohm-m. (B in Fig. 32) appears at the southwestern edge of the mapped area, possibly extending beyond the map.

Source 77

An area with apparent resistivity values of less than 20 ohm-m. (Zone A in Figure 33) appears around the source. This area is not clearly defined due to the absence of adequate coverage near the source. From the source, the resistivity values increae in all directions, with the highest values observed at the southeastern and northwestern edges of the mapped area.

Source 78

An elongated area with apparent resistivity values of about 20 ohm-m. (Zone A in Fig. 34) appears adjacent to the source. Another area with resistivity values of less than 20 ohm-m. (Zone B in Fig. 34) is observed at the western edge of the mapped area.

The resistivity values, in general, increase from the source in all directions.

Source 79

An elongated north-south trending zone with apparent resistivity of less than 20 ohm-m. (Zone A in Fig. 35), surrounds the source and is aligned with it. Apparent resistivity values increase towards the west and east from this area.

High apparent resistivities of a few hundred ohm-m. appear at the northeastern and northwestern borderline of the map.

Source 80

A one point relatively low resistivity area (Zone A in Fig. 36) with apparent resistivity of less than 20 ohm-m. appears about 1 km. north of the source. This zone possibly extends beyond the mapped area.

Another one-point relative low apparent resistivity area (Zone B in Fig. 36) is found at the eastern edge of the mapped area. This zone may also extend beyond the surveyed area.

The resistivity values increase from the source in the northeast and northwest directions and attain values as high as a few hundred ohm-m. Directly west of the source the resistivity values increase and attain a maximum value of 128 ohm-m., but decrease further to the west.

5.2 Dipole-Dipole Profiles

Five dipole-dipole profiles were measured in the area (Fig. 1).

Profile no. 1 Dipole-dipole profile no. 1 (Fig. 37) was designed to cross the long axis of the Yizreel and Harod valleys. Despite the facts that it is very long, (about 45 km.) and that it crosses a populated area, it is mostly continuous. In places where the profile is not continuous, or where turns of over 15° have been made, corrections were applied to the data.

In general, this resistivity profile can be divided into two major zones; The eastern zone, which crosses through the Harod valley, has typically apparent resistivities of over 35 ohm-m., except for a number of near surface zones with lower resistivities. However, even these lower resistivities generally measure over 15 ohm-m. The western zone, which includes most of the Yizreel valley, is characterized by apparent resistivities of less than 20 ohm-m., except for two high resistivity areas. The transition between the two major zones occurs near Merhavia (points 47-50), and is rather rapid.

The change in apparent resistivity with increasing N is not a uniform phenomena throughout the section. Two areas of increased apparent resistivity with N are defined; one between points 12 and 25, and the other between points 32 and 56. In the rest of the section, the apparent resistivity either decrease with increased N (note points 1-10), or do not show a clear change in values (points 69-77). Two areas of relatively low apparent resistivity sections are indicated. The first is at the eastern edge of the profile at the border between the Harod and Bet She'an valleys (points 1-7). Apparent resistivities at $N > 3$ in this area are mostly less than 15 ohm-m. The other area is in the Yizreel valley (points 79-82) with resistivities of less than 5 ohm-m.

Two marked high resistivity areas are observed in the Yizreel valley, one between points 61 and 66 and the other between points 87 and 89.

A gradual decrease in apparent resistivities in the Yizreel valley towards the northeast may be present, but the evidence is inconclusive.

Profile no. 2 Dipole-dipole profile no. 2 is located along the southwestern border of the Yizreel valley, in the northwest-southeast direction (Fig. 1). In the northern portion of the section, the apparent resistivity values indicate the presence of a few local highs and lows (Fig. 38). One such low is centered around point 16 and dips to the northwest. Another low appears below points 21 to 23.

South of point 31 the results indicate an extremely variable section with rapid resistivity changes possibly related to the appearance of volcanic rocks in this area. Two major highs exist centered between points 31-36 and 45-48. Between these highs appears a zone of relatively low apparent resistivity, centered around point 41.

Profile No. 3 This profile was conducted along the road from Afula to the southwest (Fig. 1). Apparent resistivity values are mostly less than 15 ohm-m. (Fig. 39). A low seeems to center beneath point 15 of this section, with resistivities of about 9 ohm-m. High resistivity areas such as those observed in the previous profiles do not exist here.

Profiles 4 and 5 Profiles 4 and 5, oriented north-south are aligned to form a traverse across the Yizreel valley (Fig. 1). The presence of the town of Afula within the line caused a gap between the two profiles. At the southern portion of the valley, between points 12 to 31 of section 5 (Fig 40), apparent resistivities are mostly over 50 ohm-m. In contrast, moving from point 12 in section 5 northward, including section 4, the apparent resistivities are mostly under 40 ohm-m.

In parts of the profiles which seem relatively undisturbed, the apparent resistivity increase with increased N. A prominent high apparent resistivity zone appears between points 11 to 16 of section 5, and resistivities at N-8 reach over 200 ohm-m. At the northern edge of section 5 and near the town of Afula, apparent resistivities seem to significantly decrease. However, the lack of measurements in the area around the town prevents a complete evaluation of this anomaly. North of the town of Afula, in section 4, the resistivity section is not uniform. Most values are between 15 and 40 ohm-m. in similarity with points 4-11, of section 5. In general, the apparent resistivity in this area increases with depth.

5.3 Depth Soundings A total of 60 modified Schlumberger and monopole depth soundings, two for each source (Figure 1) were measured in the present survey. The results, presented separately for each source, are shown in the Appendix.

In general, the soundings could not be used in interpretation. The few which were interpreted indicate a highly resistive layer with resistivities of 90-200 ohm-m. under the cover of a low resistivity layer with resistivities of 8-20 ohm-m. The highly resistive layer was found at depths of 100-220 m.

Three soundings indicate that the second layer has resistivities in the range of 14-35 ohm-m.

6.0 DISCUSSION

This survey, which included some 34 roving dipole sources, close to 80 km. of dipole-dipole profiling and a number of depth soundings, provides good coverage of the Yizreel and Harod valleys, and nearby areas. In addition, a large number of earlier depth soundings (Levanon, 1964; Levanon, 1976; Levanon and Barner, 1970; Paparo, 1974) supplied additional

information. The discussion which follows will attempt to reach conclusions concerning the possible existence of hot water systems in the area, using all the available information. It is not the aim of this report, however, to make a complete structural and lithological evaluation of the valleys from this data. The geothermally related analysis will be made through trying to establish the normal appearance of the apparent resistivity on the maps and profiles in the various portions of the area and then by identifying areas of anomalous low resistivity.

A generalized resistivity section in the area will include a few meters of near surface soil and alluvial coverage, with true resistivities of 3 to 10 ohm-m. In the Harod valley and nearby areas, the upper and lower basalts appear below the soil, typically with resistivities of 20-165 ohm-m. The basalts extend to near Afula on the surface, but it is obvious from the new data that it extends further to the west in the subsurface. In addition, a northeast trending line of exposed basalts is found west of Afula (Fig. 1). The extensions of the high resistivity basalt layer beyond the exposed areas, as evidenced by the data, can also be seen in some of the drillholes (Gid'on 1-5). Some basalt seems to exist, in the subsurface under much of the area, including some spots where no indication is given by the resistivity survey.

In places where the basalt is weathered, thin or missing, the soil is most probably underlain by the Eocene chalk and limestone unit and by the Senonian chalk which usually have resistivities of 20-100 ohm-m. Still lower appears the Turonian, Cretaceous and Jurassic limestone sequence with resistivities of 100-250 ohm-m. In places where the basalt is clearly found, it is usually underlain by the same limestone complex. Basement was not found in any of the drillholes in the area and depth to the basement is not known, but is probably a few kilometers deep.

In general, where near surface resistivities are low and resistivities at depth are higher, roving dipole apparent resistivity maps

are expected to show an increase in apparent resistivity values with increased distance from the source. This common phenomenon is caused by the accumulating affect of the high resistivity layer at depth. At great distances, the apparent resistivity might approach asymptotic values representing the true resistivities of the second layer. In the same manner, in the dipole-dipole profiles apparent resistivities are expected to increase with increased N.

However, the complexity of the geological section and the large lateral variations in true resistivities due to the appearance of the volcanic flows, largely complicate this schematic picture. For example, in cases where the source in the roving dipole survey is located over high resistivity volcanics, apparent resistivity values are high near the source and may remain relatively high throughout the map. Also, if the source is located over volcanic rocks while the receivers are placed over a sequence which is mostly devoid of the high resistivity volcanics, a low apparent resistivity closure may develop in this area. A good example of this process can be seen in Figs. 17-18, which show roving dipole data around sources 57-58. The two sources are located over thick basalts and apparent resistivities near the source are high and do not increase regularly with distance, but depend instead on the variations in lithology. In particular, the reduction in thickness and possible alteration of the volcanic layer causes a marked decrease in apparent resistivity values.

Low apparent resistivity areas which have been defined in the preceding sections have been compiled in two maps. One, (Fig. 41) is the compilation of the dipole-dipole data at N=4 from all five sections. The other, (Fig. 42), which is called the interpretation map here, is the compilation of all low apparent resistivity areas which have been defined on the various dipole maps. These, of course, were all low apparent resistivity areas, relative to their surroundings. Many of the areas which were described (possibly all of them) could be explained by means other than the existence of high temperature reservoirs in the subsurface.

Many are found to be the result of the specific locations in which the sources and receivers were placed in regard to each other, and therefore, they are expected to disappear when illuminated by another source or by a dipole-dipole section. For example, low resistivity areas immediately surrounding the source reflect only shallow current penetration as described above. In such cases, these low resistivity areas will appear normal when illuminated from a distant source.

A typical appearance of such a shallow near source anomaly is an ellipse whose long axis is aligned with the source, relfecting the shallower penetration along the source direction as compared with that perpendicular to the source.

Another such misleading low resistivity area will appear on the roving dipole maps at receiver stations where no volcanics are present, if the source is situated directly over shallow volcanics. In this case as well, a distant source located in a non-volcanic area will eliminate the area as an anomalous zone.

The criterions used in this analysis for the acceptance of an apparent resistivity low as true resistivity anomaly, and as a possible geothermal target, are as follows:

a. The apparent resistivities have to be significantly lower than those of the surrounding areas.
b. Apparent resistivity values are low in absolute terms, and preferably lower than 10 ohm-m.
c. If the area is seen as a low on one map, it should also be seen as a low on all other maps and cross-sections which include it.
d. There are no other geologically straigth forward non-geothermal mechanism to explain the resistivity low.
e. The geothermal explanation of an anomaly is in line with all other available information and in particular with water temperatures in springs and drillholes.

Another point which should be made before going into a detailed discussion of the various anomalies is the question of the reliability of dipole-dipole profile no. 2, (Fig. 38) in the context of this survey. This profile, which was made along the Tiveon-Megiddo road, is placed near the fault system separating the valley and the elevated block bordering it on the southwest. Therefore, the profile, which is calculated as though it was conducted in a one dimensional medium, is distorted. However, since in this survey our interest lies mainly in delineating relative anomalies rather than absolute ones, the profile is expected to include valuable and relevant information despite the obvious distortion.

The western part of the Yizreel valley is an area in which the resistivity offers no indications to the presence of volcanic rocks. Apparent resistivities are typically less than 40 ohm-m. both in the roving dipole maps and in the dipole-dipole sections. However, some volcanic rocks may exist without being apparent on the resistivity maps and cross-sections.

As can be clearly seen, in Fig. 42, most of the zones in the western portion of the Yizreel valley seem to center around Yoqne'am. The meaning of the clustering of low resistivity zones in this area is that the area must be regarded as a lower resistivity area, regardless of source location and orientation. This, in turn, probably means that the area may be expected to have lower true resistivities than the surrounding areas at the same depths.

The same area has also been traversed with the dipole-dipole profile no. 2 (Fig. 38, points 13-20). It can also be seen that it includes lower apparent resistivity values even though the area does not appear uniformly low. In general, it seems that the region near Yoqne'am with limits which are not clearly defined, is an area of low resistivities.

The source of this low is not clear. The existence of a hot water reservoir at the subsurface can be expected to produce an anomaly of this sort. However, more conventional explanations are more likely. One such explanation concerns the subsurface distribution of the high resistivity volcanics. If the volcanics which are known to exist to the north of this area (Alonim drillhole Fig. 42) and to the southeast of it, (Mishmar Haemeq drillhole Fig. 42), do not exist in this particular area, the result will be a low resistivity anomaly. An examination of the magnetic map (Folkman and Yuval, 1976) shows that a low magnetic signature exists over the area, possibly indicating the absence of the volcanic rocks. Similarly, in the gravity map (Ginzburg, 1960) a marked low centered near Ramat David, includes this area, and may be related to the distribution of the dense volcanic rocks. A water well which was drilled immediately west of this area (Yoqne'am Well, Fig. 42) did not encounter any basalt and seem to sustain this interpretation.

An alternative explanation which is non-geothermal,and is in accordance with the magnetic and gravity data, assumes the presence of a structural low in the Yoqne'am area with the higher density rocks of the Cretaceous and Jurassic age at greater depth.

The determination of the appropriate explanation will depend on some drillhole data to be acquired from the center of this area and on data concerning water temperatures at the subsurface.

Other low resistivity zones which have been observed in the western portion of the Yizreel valley are also illustrated on the interpretation map. However, none of those anomalies remained as such when viewed with a different source. In addition, Zones 72-A and 42-B which are shown as lows are near source lows and probably are not significant anomalies, unless additional supporting data pointing to that are found.

Dipole-dipole profile no. 1 (Fig. 37) and the dipole-dipole composite map (fig. 41) show that the area northwest of Kefar Barukh has low resistivities, but the area of roving dipole 42-B in particular does not indicate any outstandingly low values. The appearance of low values in the dipole section is most likely to be explained by the termination of the volcanic rocks which are known from the Kefar Barukh lake. The gravity map (Ginzburg, 1960) does indicate that this resistivity low and the low which was defined near Yoqne'am are parts of the same structure.

Special attention should be paid to some very low apparent resistivities observed by dipole 81-82 in profile no. 1 (Fig. 37). The observed values of less than 1 ohm-m., (corrected for the noncotinuity of the profile) are much lower than those observed in any other part of the surveyed area. The fact that the exceptionally low values are observed by only one dipole may indicate that they are the result of some very local dipole conditions. Although there is no conclusive proof that this is the case, the localization of the zone almost eliminates the likelihood that the anomaly is geothermally related.

Further to the east, a line of surface volcanics is seen crossing the Yizreel valley trending from northeast to southwest in the vicinity of the Kfar Barukh lake. For the most part, these are clearly identified in the various dipole-dipole sections and composite dipole-dipole map (Fig. 41) as well as in the roving dipole maps, as areas of high apparent resistivity. Exceptions to this general trend are two local roving dipole lows (54-A and 53-A, Fig. 42) in the vicinity of Hayogev. Each of these lows is located near a source, and as such do not indicate, in all likelihood, any geotermally related anomalies.

Additional evidence for this interpretation can be found in the fact that these lows do not appear as such when illuminated by distant sources nos. 41 and 42.

The fact that low apparent resistivity values are measured near sources which are located on a surface exposure of basalt can probably be explained by a local change in the resistivity of this rock. The Gid'on 3 well, located near source 54, penetrated into weathered basalt and then into a generally chalky section (as indicated by the composite log). This weathering of the basalt is likely to cause the general section to include lower resistivity basalt and chalk over higher resistivity limestone. A combination of this sort will tend to form low apparent resistivity closure in the vicinity of the sources.

A few low apparent resistivity zones are indicated in the interpretation map (Fig. 42) directly west and north of the town of Afula. Of these, zones 54-C, 44-A, 43-A and part of 48-A seem to coincide. Directly south of this cluster of zones, dipole-dipole traverse no. 3 (points 8-18, Fig. 39) indicates an apparent resistivity somewhat lower than that found to the southwest. However, the difference is not a marked one, and apparent resistivities of over 10 ohm-m. are probably not sufficiently low so as to be of geothermal interest. Dipole-dipole no. 1, situated directly north of this cluster, does not indicate the presence of an anomalous zone either. Near surface apparent resistivities in the vicinity of points 57-60 seem to be lower than those measured to the southeast, but they seem to be part of a general trend of decreased resistivity in the northwest. Sharp increases in resistivities beyond point 60 are probably related to the existence of volcanics in the shallow subsurface.

It is difficult to reach a conclusion concerning the nature of the observed cluster of lower apparent resistivity zones. Over the same area and to the southwest, a low apparent reissitivity is observed in the composite dipole-dipole map (Fig. 41). In addition, a low appears in the magnetic map (Folkman and Yuval, 1976) and extends to the south. The area is a part of a larger low gravity anomaly which extends to the east (Ginzburg, 1960).

Although the existence of a geothermal source can not be ruled out entirely, it is most likely that the cluster of low resistivity zones west of Afula is related to change in basalt thickness and character change in water salinity or a combination of the two. Evidence to this effect can be found in the Sarid 1 and Gid'on 5 wells (Fig. 42) which penetrated a section of alternating basalt, weathered basalt and fossiled soil unlike the more continuous basaltic section found in wells outside this area. Saline water were found in the Sarid 1 well (Rosenthal,personal communication).

In addition one must consider the factor that two Schlumberger soundings which were conducted in the area (Levanon, 1965) clearly indicate that true resistivities in the subsurface are high.

Two low apparent resistivity zones, 49-A and 50-A, which are situated south of the town of Afula, are both near source zones. Dipole-dipole profile no. 5 (Fig. 40) which crosses the area indicates what seems to be a normal section in the absence of a significant layer of high resistivity volcanics.

The southern-most part of the Yizreel valley, the Taanach district, does not show any anomaly of geothermal interest. The low apparent resistivity closures which appear near Nir-Yafe (47-A, 48-A) and Avital (49-A, 50-A) are near source closures with no additional support.

East of the town Afula and north of the Harod valley, most of the area included in this survey is covered by surface basalt flows (Fig. 1). At places, the basalt is eroded and Eocene is exposed.

Resistivities in this region are high as expected in an area which is covered extensively by basalt, similarly to the high apparent resistivities recorded on the volcanic flows in the center of the Yizreel valley. Apparent resistivities seem to decrease with distance, as expected from a section in which the uppermost layer has higher resistivities than the

layers beneath it. An example of this distribution can be seen in the maps of sources 67-68. A different pattern can be seen around sources 77,78,79 and 80, where low apparent resistivity closures appear in the immediate vicinity of the sources. This is a pattern which is generally expected where the surface layer has lower resistivity than the layers underneath it. The explanation for the existence of this pattern probably lies in the fact that surface volcanics are very thin or altered and the transversal resistivity is not great enough to significantly influence the results. However, the four low apparent resistivity closures around these sources do not seem to have any additional support from distant sources and do not have any geothermal significance.

Four additional low apparent resistivity closures appear on the interpretation map; 73-A, 75-A, 76-A and 78-B. Of these, zones 75-A and 76-A are near source and are not seen as low resistivity zones when illuminated from sources 73 and 74. Zones 73-A and 78-B are small and do not have any additional support, and are, therefore, discarded. Zone 80-A, in the eastern point of the Isachaar heights, is a one point zone on the edge of a major fault, and in all probability is not significant.

Directly north of the Harod valley and on its northern slopes, a number of low apparent resistivity zones have been described and shown on the interpretation map, but none could be proven to be of geothermal significance.

Near En Harod and Tel Yosef, zones 59-A, 58-A and 59-B seem to form a trend which is most likely to be related to the surfacing of the lower resistivity Eocene sequence in the area and the erosion of the volcanics.

Of interest are a series of low apparent resistivity zones north of Sede Nahum, namely zones 79-A, 63-C, 61-B, 63-A and 62-A. Their appearance is not completely explicable at this time. It will be noted, however, that for the most part they do not coincide and therefore there is no one outstanding portion of the area which can be singled out as a low resistivity area. As there is no clear evidence pointing to the existence of low resistivities in this area, it should be disregarded as a target for additional research.

Another likely explanation for the appearance of many of these lows concerns the existance of saline water in the subsurface. Rosenthal (Personal communication) found saline water in the areas covered by anomalies 61-A, 59-B, 58-A, 77-A, 59-A, 78-B.

In the Harod valley a few low apparent resistivity zones appear to form two clusters, one near Gid'ona and one about 2 km. to the east. Two additional zones, 50-B and 50-C, are small and without supporting evidence, and therefore will not be considered further. The valley is also traversed by dipole-dipole profile No. 1. In the area under consideration, apparent resistivity values are clearly high. Hoewever, the two clusters mentioned above are not directly traversed by the profile and it is possible that the profile does not depict the situation south of it. A closer examination of the cluster near Gid'ona shows that it is centered around a water reservoir which is expected to reduce resistivities. In addition, one of the low resistivity zones, 57-A, may reflect the fact that the source is located over thick surface volcanics, while the receivers were placed in the center of the grabens where thin volcanics interbedded in sedimentary section are present.

The other cluster is composed of Zone 60-A, which encircles source 60, and Zone 58-B with source 58, which is situated over high resistivity surface volcanics.

The two low resistivity clusters may be as well related to the presence of saline water in the subsurface although no clear indication for their existance is available.

In addition, both areas were also covered by other sources which did not indicate the existence of anomalous zones. It is likely that none of these two zones are of geothermal significance, but since the matter is not resolved some limited work will be recommended.

In the eastern-most portion of the Harod valley, close to where it borders on the Bet Shean valley, apparent resistivities drop, forming Zones 63-B and 64-A. There is little doubt that these zones reflect the transition into the more conductive section in the Bet Shean valley, rather than possessing geothermal relevance.

Special attention was given in the Introduction to En Shoqeq springs (Fig. 1). The springs which will be discussed in greater detail in another report, were also covered with sources 61 (Fig. 21) and 62 (Fig. 22) of this report. No evidence for low resistivities is observed.

7.0 CONCLUSIONS AND RECOMMENDATIONS

The present survey, which was aimed at finding areas of geothermal interest in the Yizreel and Harod valleys, did not indicate the clear presence of any geothermally related low resistivity anomalies.

The various low apparent resistivity zones observed on the roving dipole and the dipole-dipole surveys could mostly be explained by the known resistivity section and be correlated with the local geology.

Of the areas which were described in the report, only one stands out as an anomalous area which justifies addtional work. This area, near Yoqne'am in the Yizreel valley, was shown to correspond with low resistivities. Although the relatively low resistivities can be explained by non-geothermal mechanism, and in particular by the effect of the thickness and depth of the basalts in the area, this explanation is not conclusive. It is therefore recommended that some limited effort be made to establish the temperature regime in the area by drilling two 100 m. holes and measuring temperatures.

Of the other anomalies which were described, the only one which may warrent additional work is the one near Gid'ona (Fig. 42). If no water well exists in this area, it may be of interest to drill two more shallow temperature holes at this location.

In general, the results of this survey show that the area of Emeq Yizreel and Emeq Harod should not be given high priority for geothermal exploration and development.

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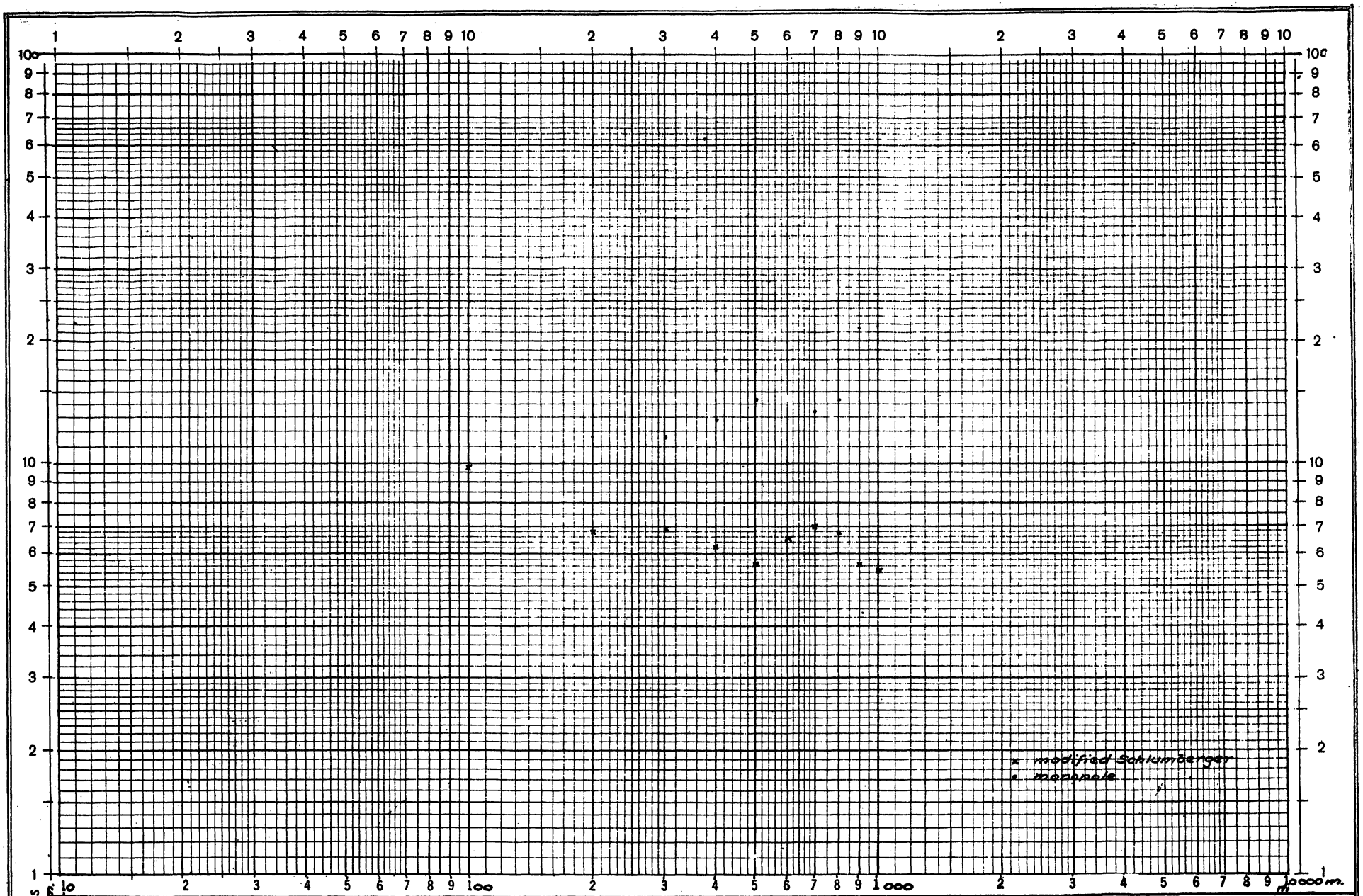
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APPENDIX

(Modified Schlumberger and Monopole Sounding Curves)



D.P. Nº 41

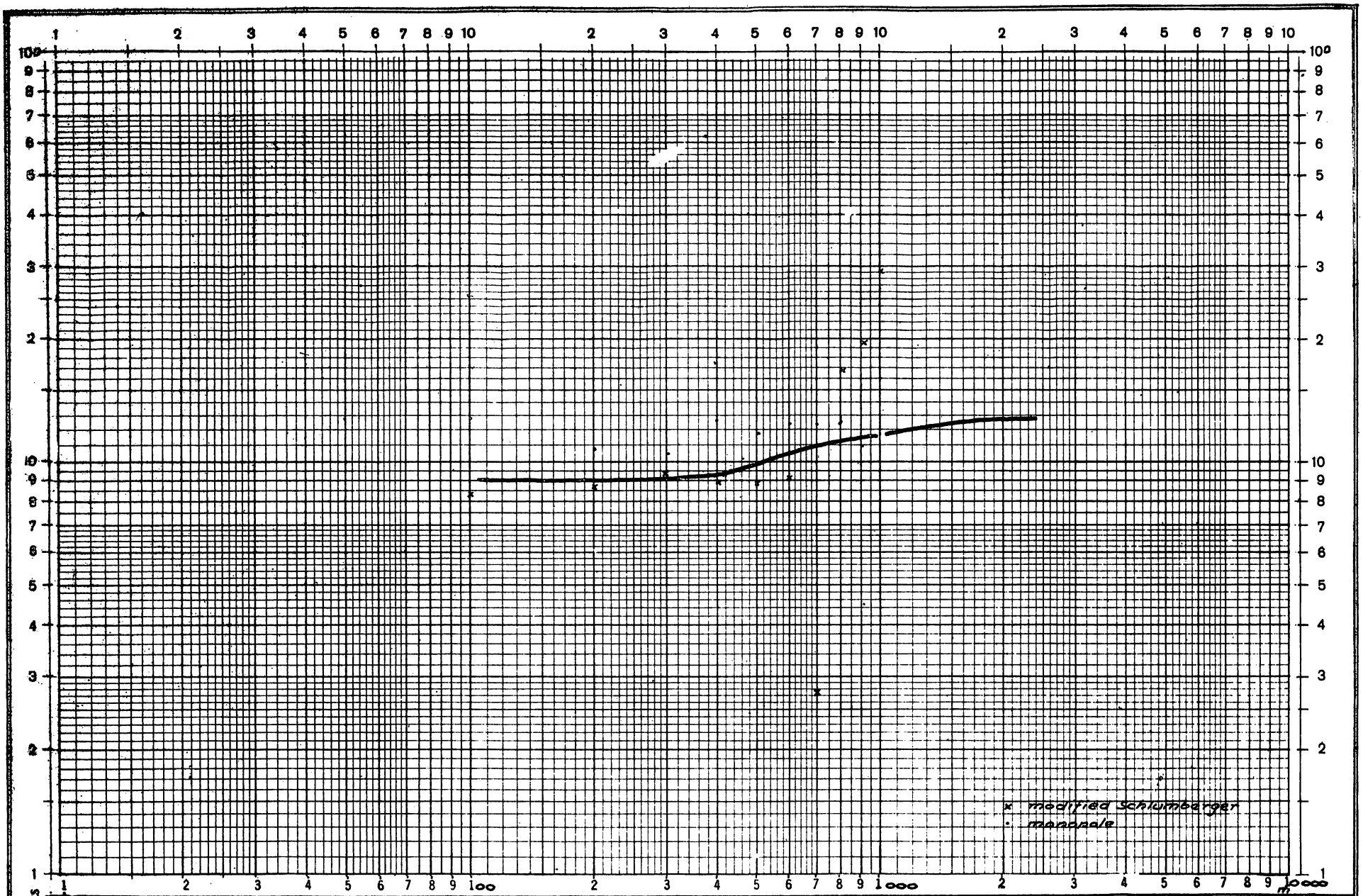
FILE Nº 001

JOB Nº 87/149/74

LOCATION : EMEQ YIZREEL

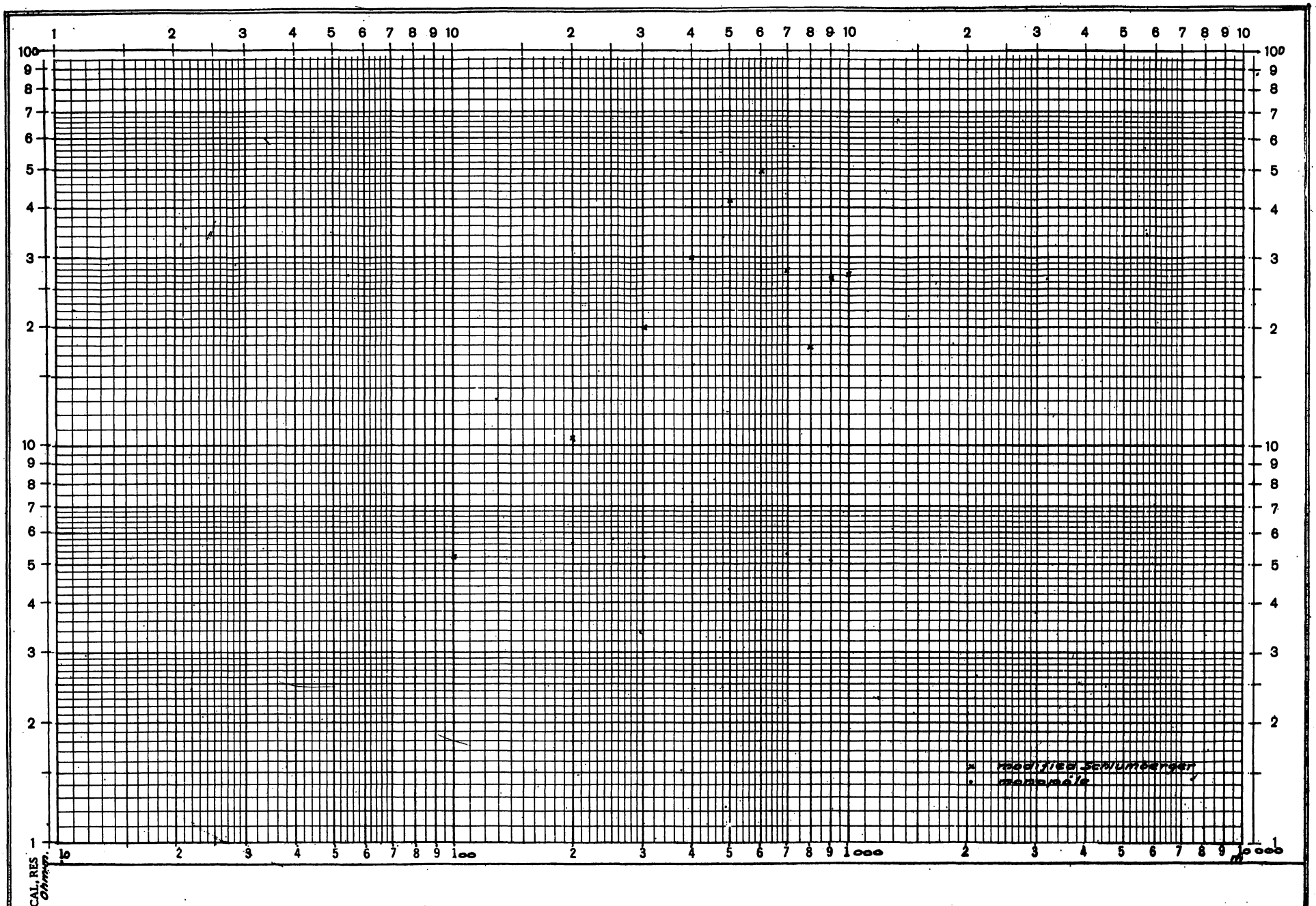
COORDINATES : 164400 - 228950 165200 - 230650 ELEVATION :

(54076) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS



D.P. Nº . . 42.	LOCATION :	EMEQ YIZREEL
FILE Nº 001 . .	COORDINATES : 164400 - 228950	166050 - 227800 ELEVATION :
JOB Nº GT/149/74	(54077)	THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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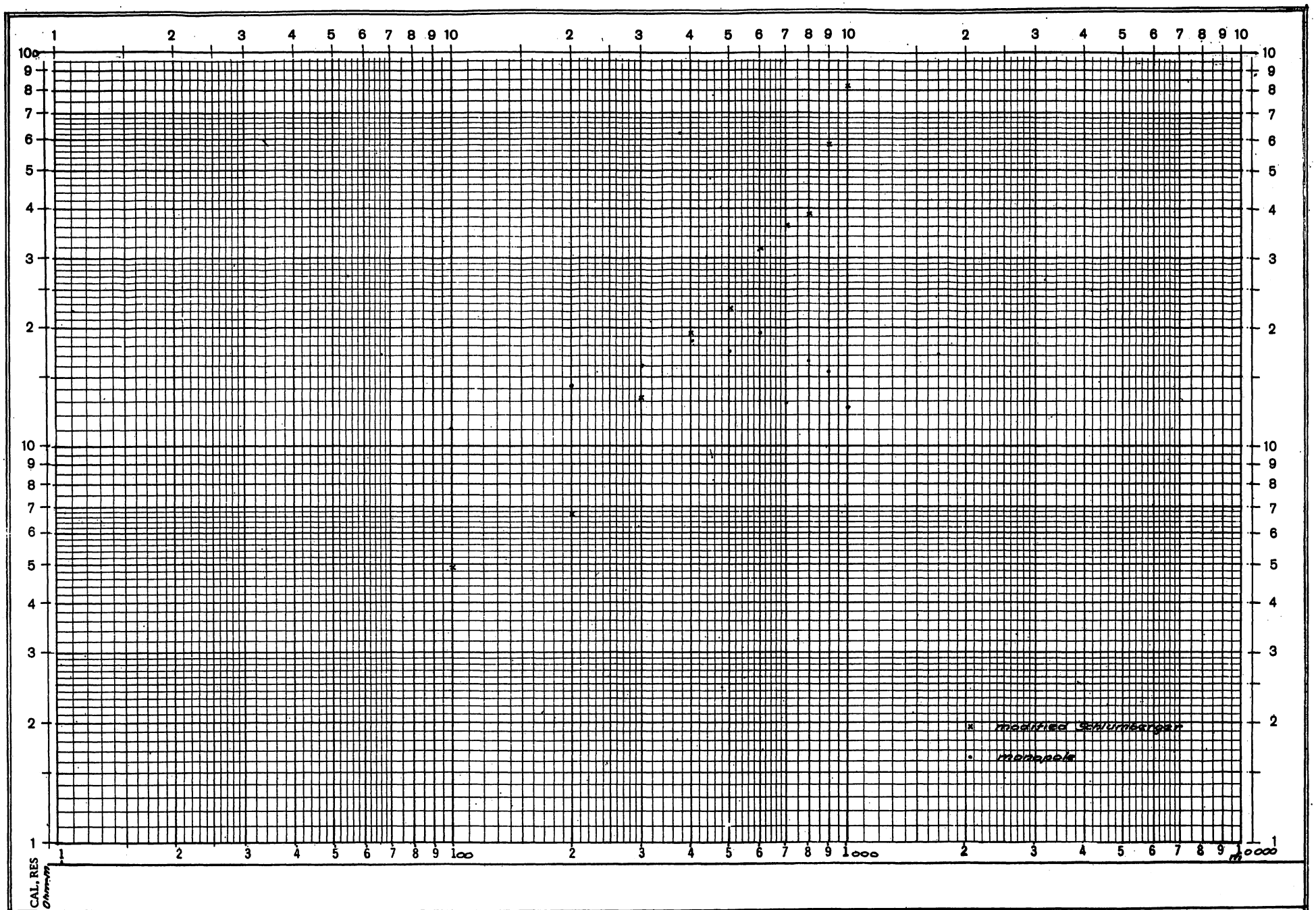


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D.P. Nº. 43.
FILE Nº 001.
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 174 550 - 287 200 175 000 - 285 150 ELEVATION :
(54078) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

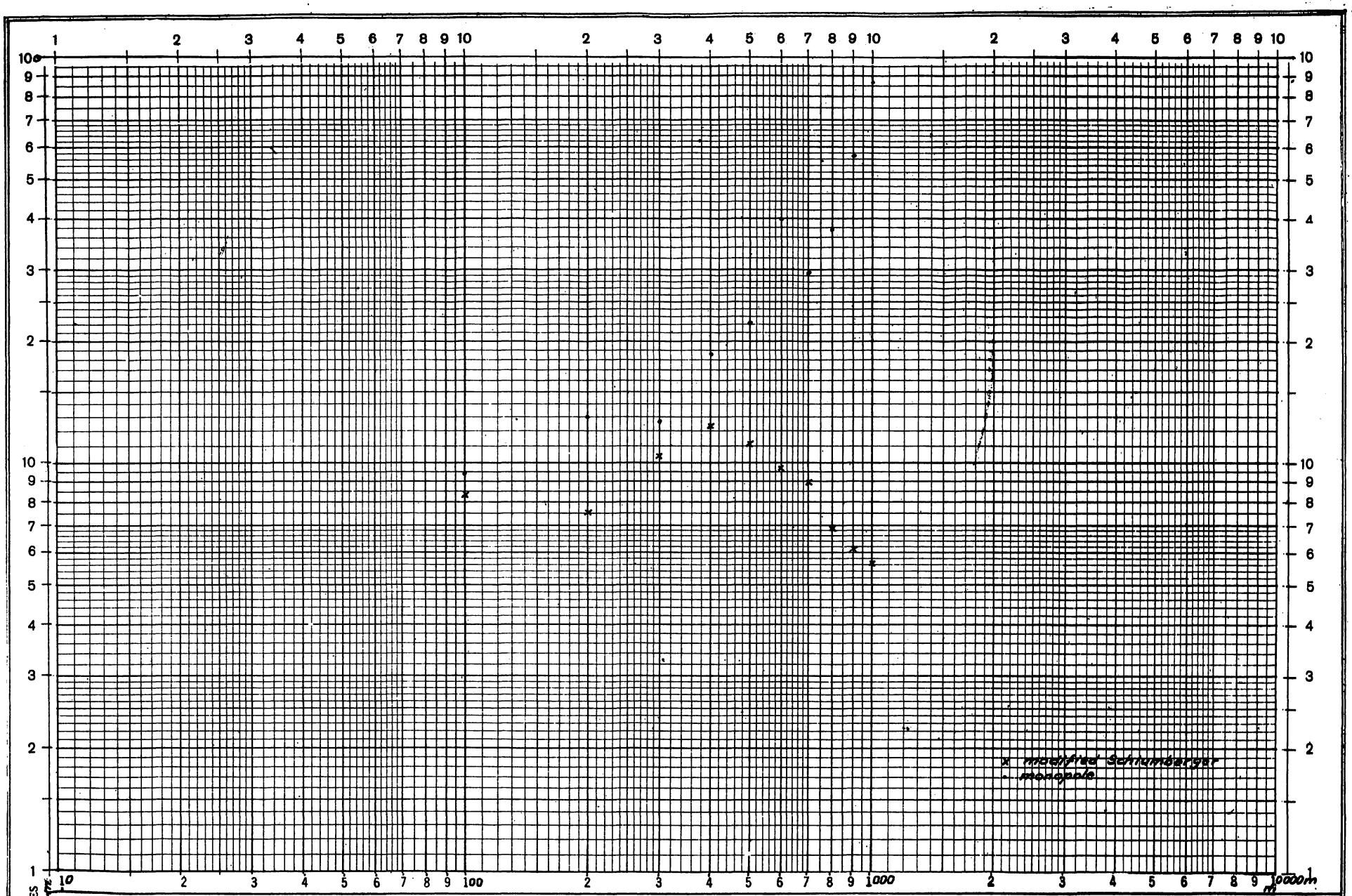
6.130



D.P. Nº 44
FILE Nº 001
JOB Nº GE/149/74

LOCATION : EMEQ - YIZREEL
COORDINATES : 174550 - 227200 176400 - 227550 ELEVATION :
(54079) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

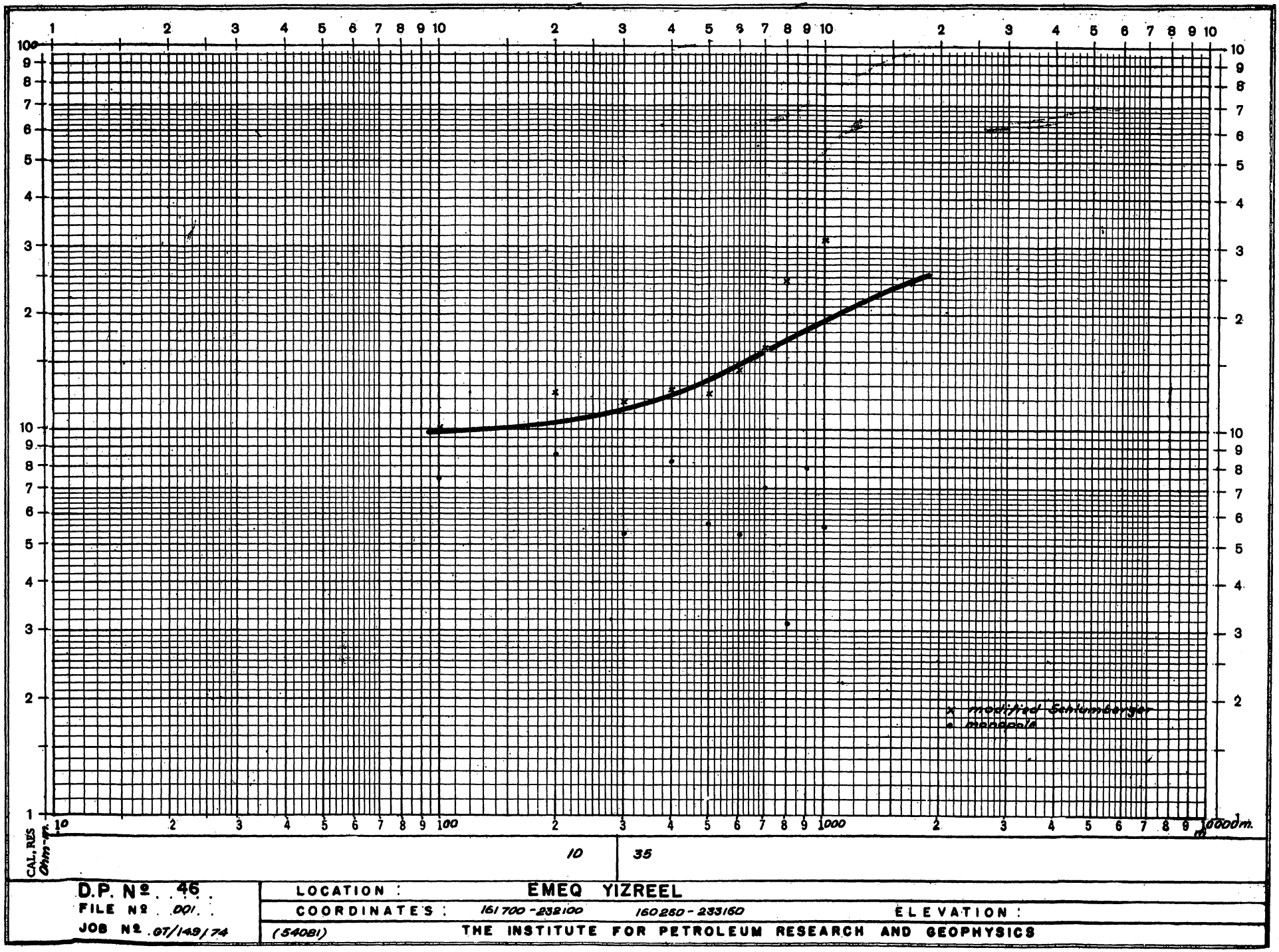
6.130



D.P. Nº 45
FILE Nº 001
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 161700-232100 162700-233600 ELEVATION :
(54080) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

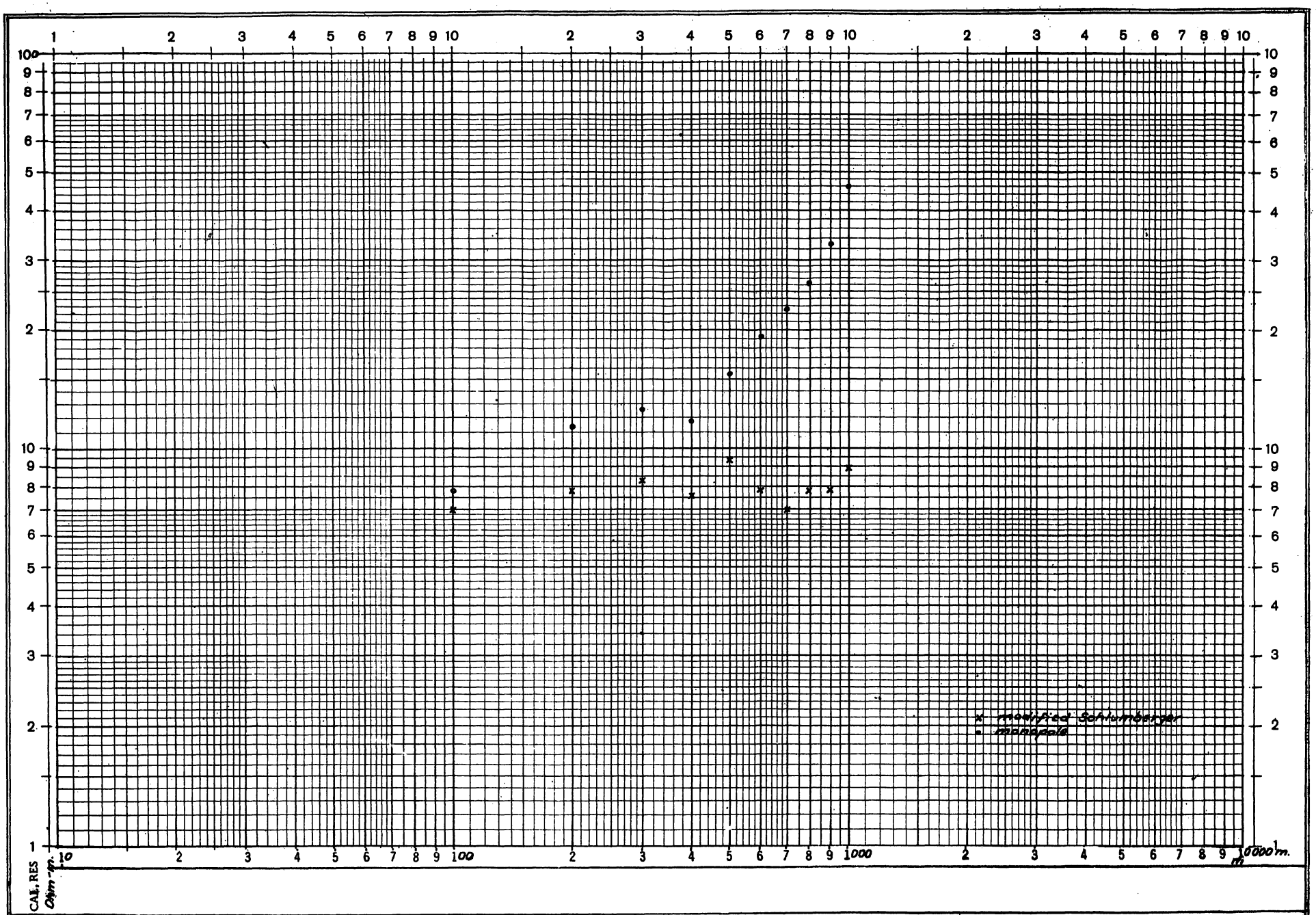
6.130



D.P. Nº 46
FILE Nº 001
JOB Nº 07/149/74

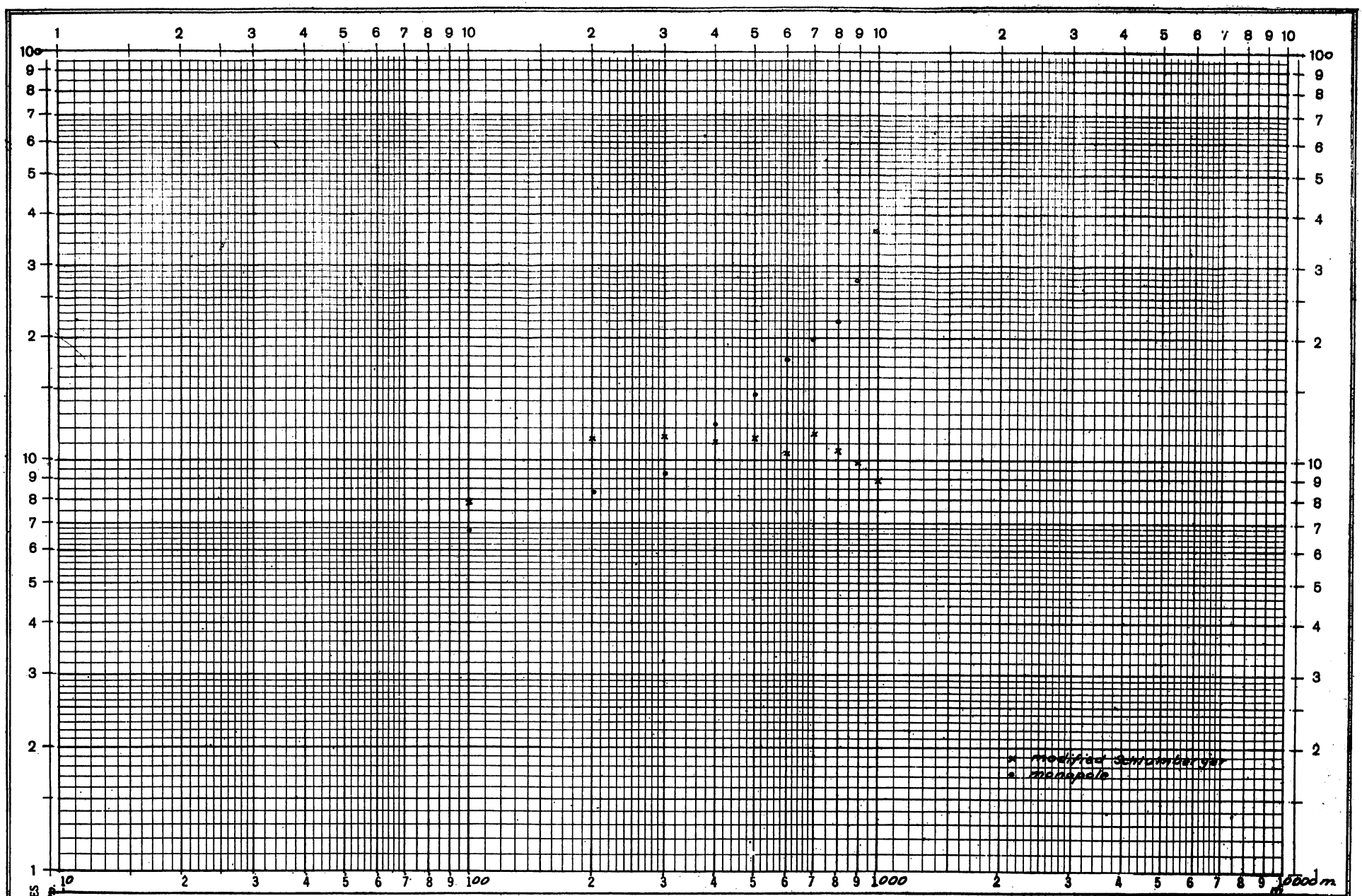
LOCATION : EMEQ YIZREEL
COORDINATES : 161700-232100 160250-233150 ELEVATION :
(54081) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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D.P. Nº 47	LOCATION : EMEQ YIZREEL		
FILE Nº 001	COORDINATES : 172300 - 219900	171200 - 221150	ELEVATION :
JOB Nº GT/149/74	(54082) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS		

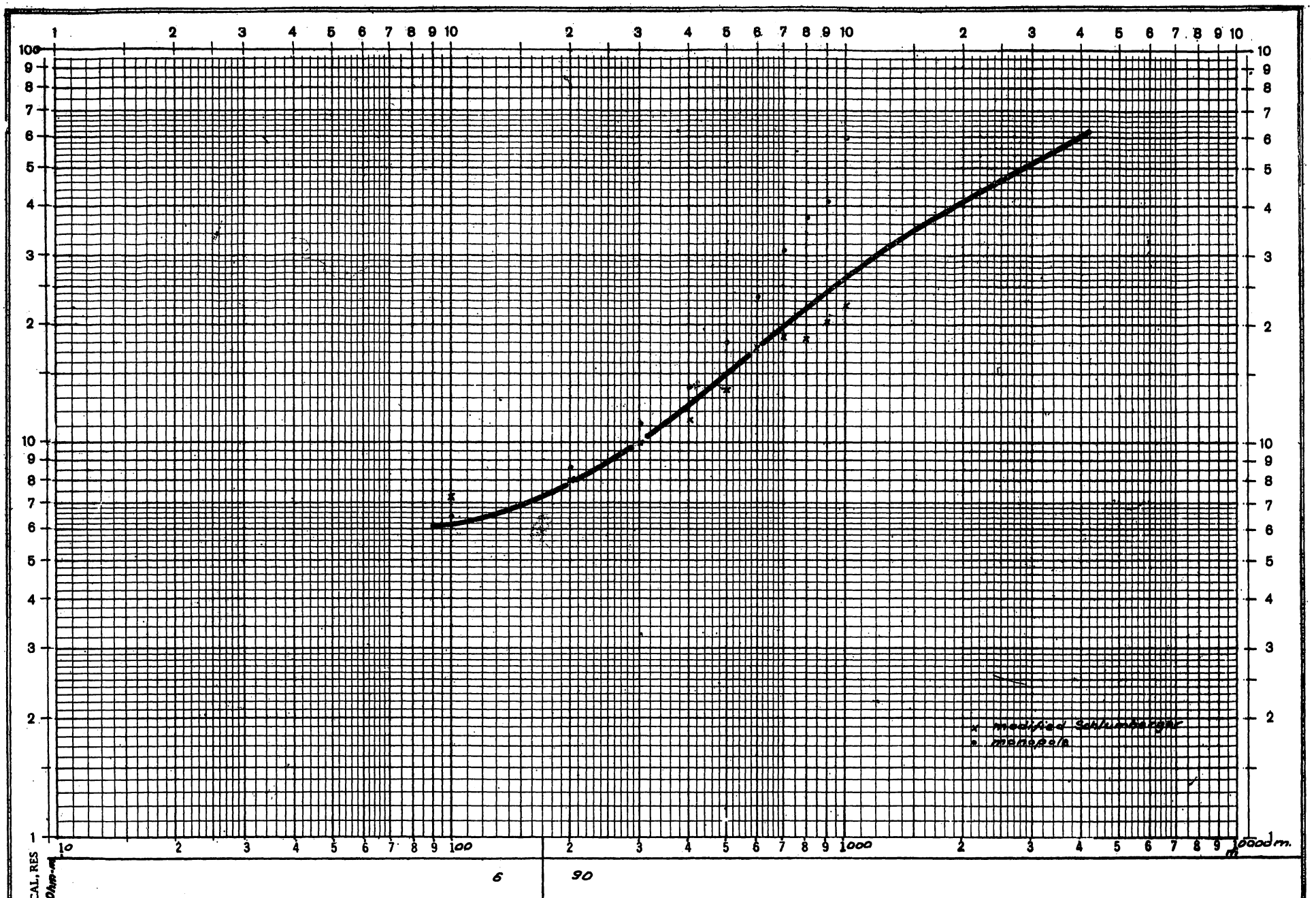
6.130



D.P. Nº. 48.
FILE Nº .001. .
JOB Nº 67/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 179300 - 219900 173450 - 221200 ELEVATION :
(54083) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

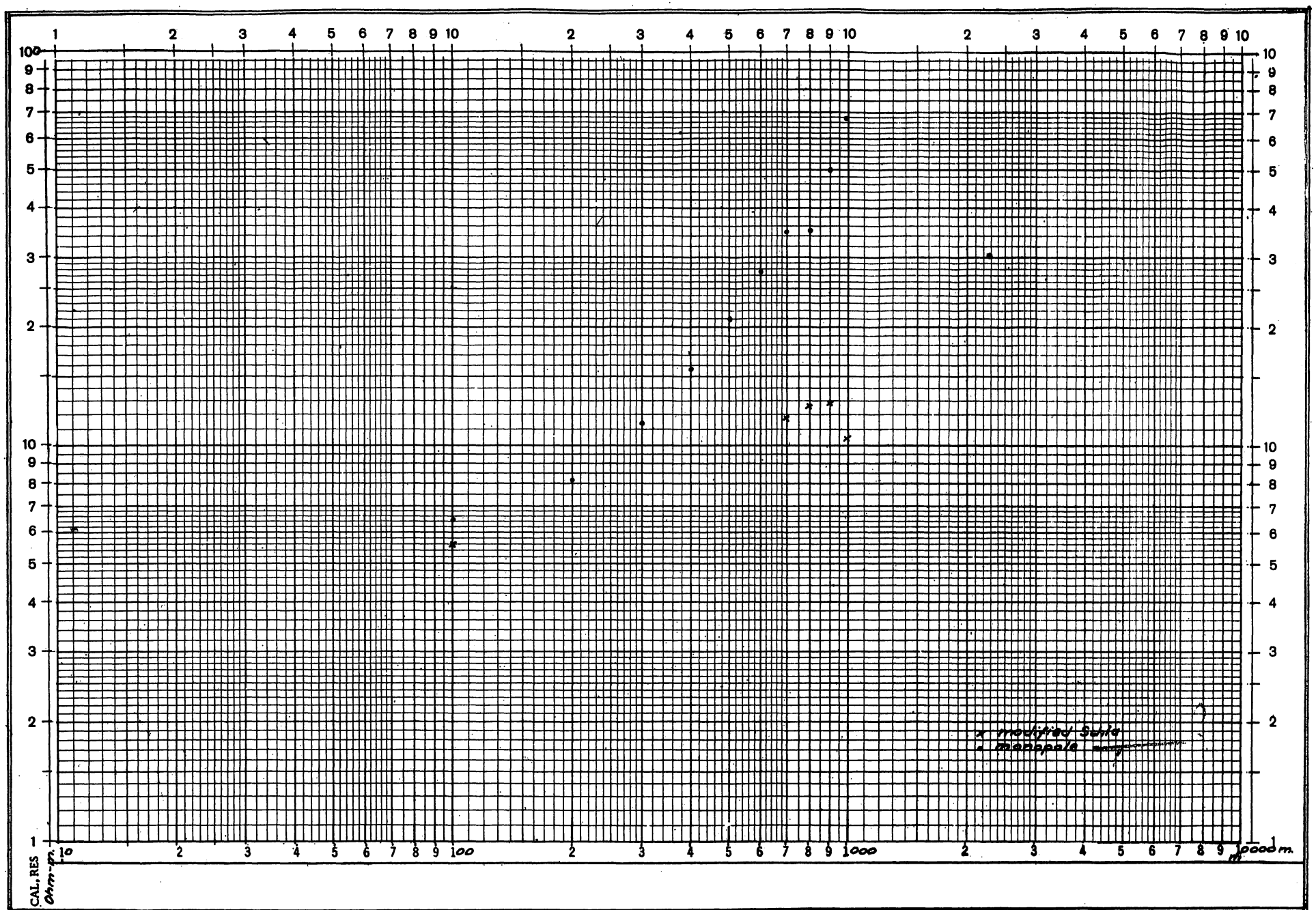
6.130



D.P. Nº. 49..
FILE Nº .001..
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 177400 - 218700 177300 - 220700 ELEVATION :
(54084) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

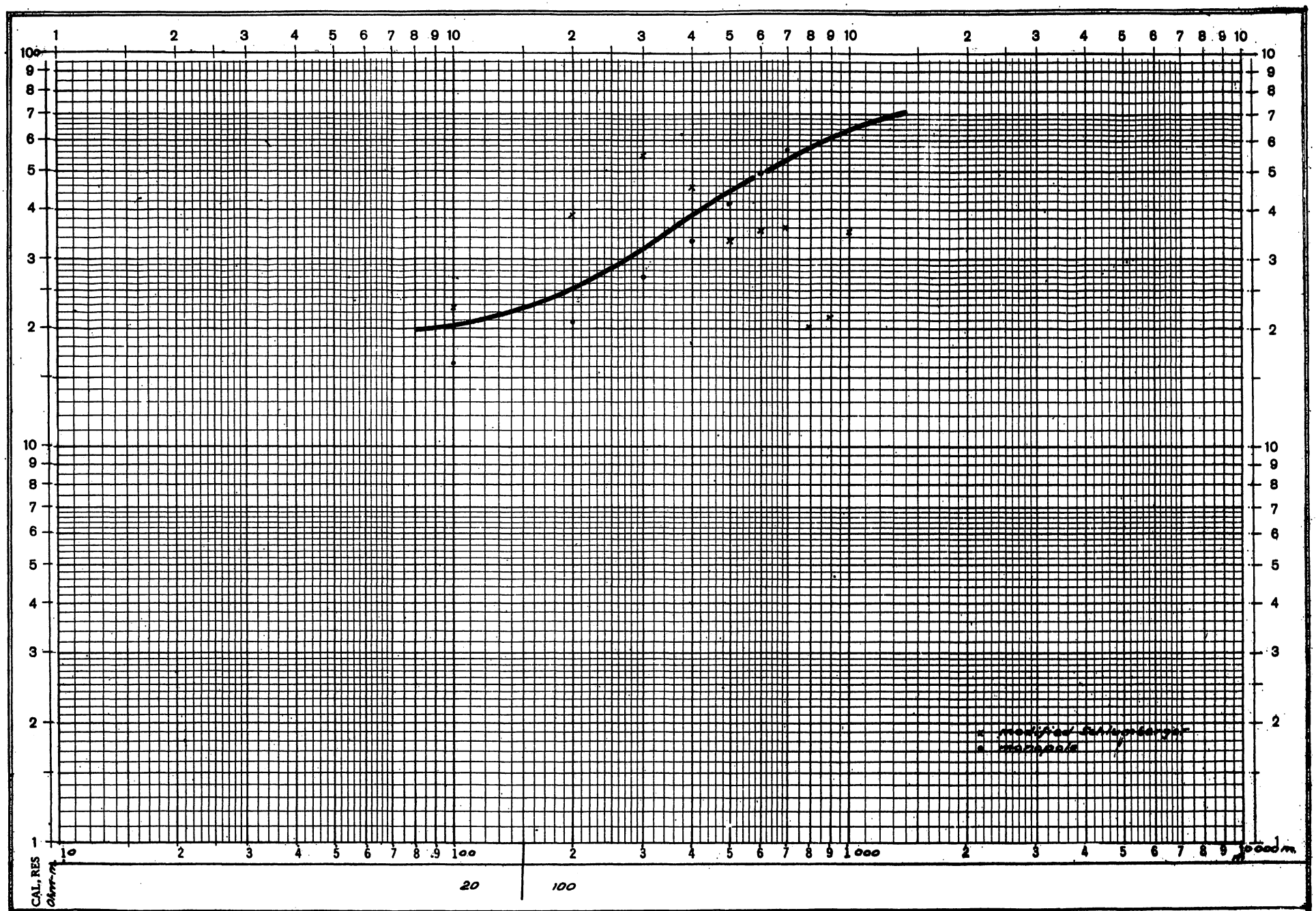
6.130

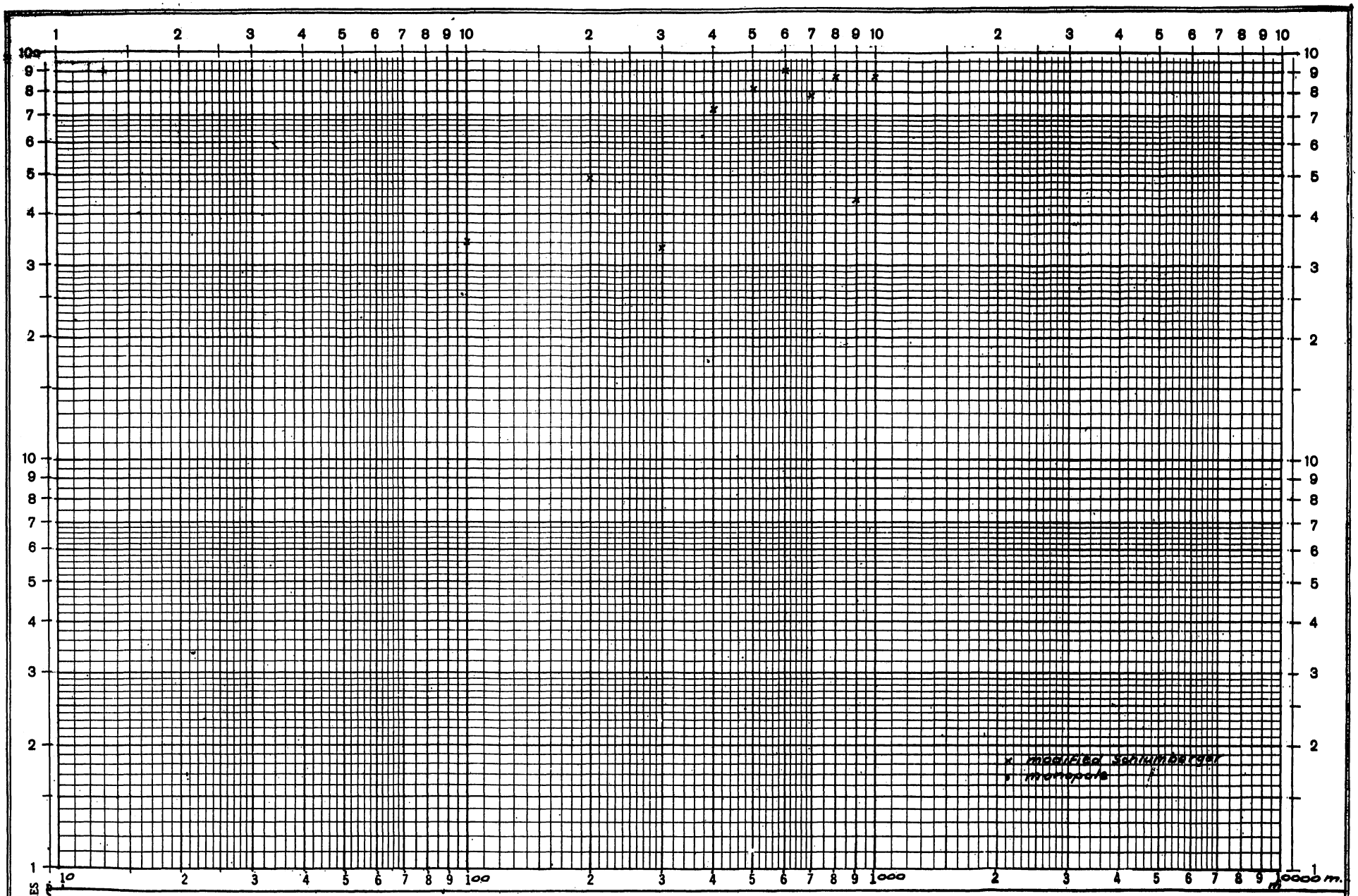


D.P. Nº . . 50
FILE Nº .001. . .
JOB Nº 87/142/74 (54086)

LOCATION : EMEQ YIZREEL
COORDINATES : 177400 - 218 700 179 100 - 218 600 ELEVATION :
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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D.P. Nº 51
FILE Nº 001
JOB Nº GT/149/74
LOCATION :
COORDINATES : 176 150 - 213 400 176 780 - 214 950 ELEVATION :
(540B5) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
6.130



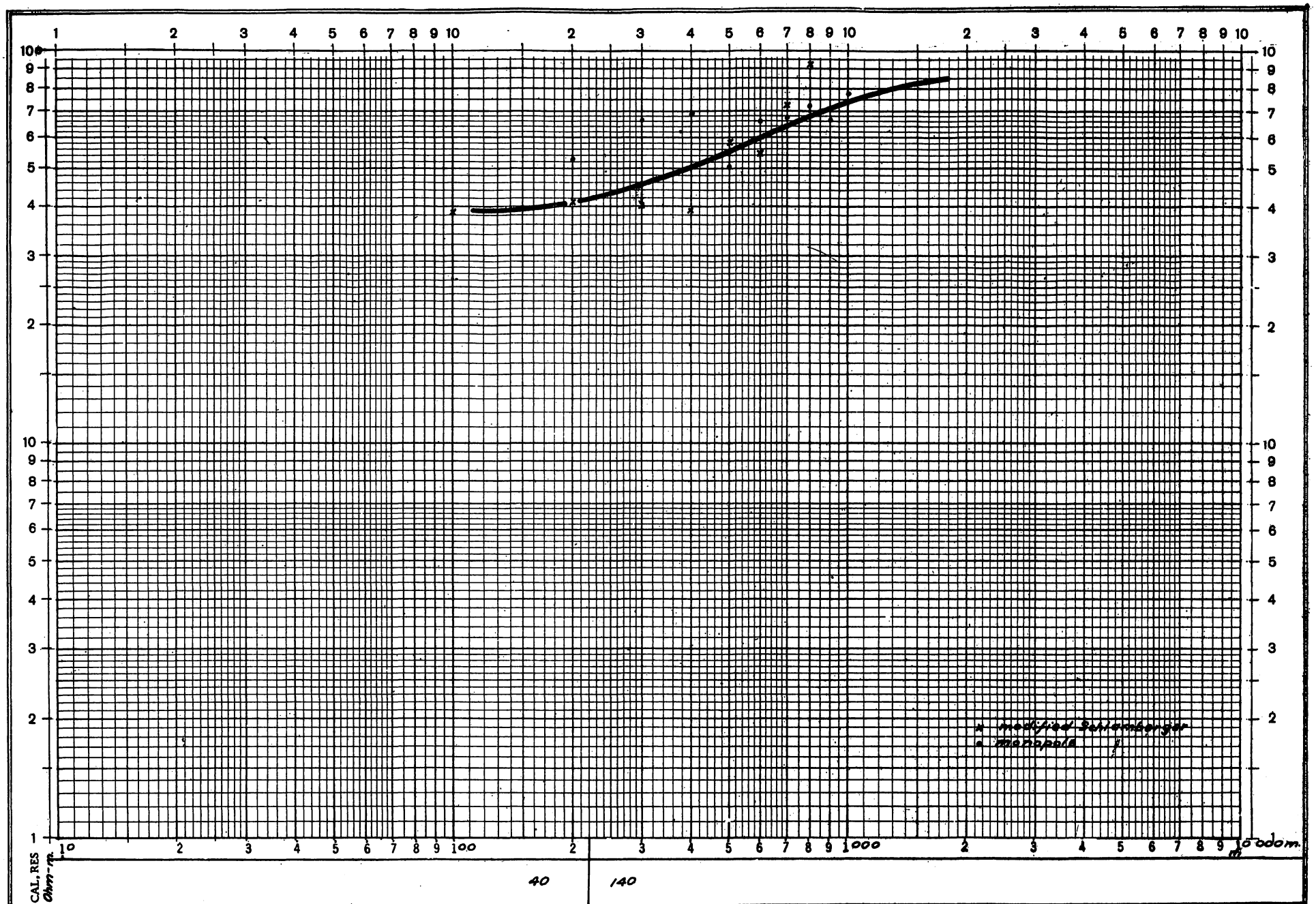
D.P. Nº . . 52.
FILE Nº . 001 . .
JOB Nº 67/149/74

LOCATION : EMEQ YIZREEL

COORDINATES : 176150 - 213400 174400 - 214100 ELEVATION :

(54067) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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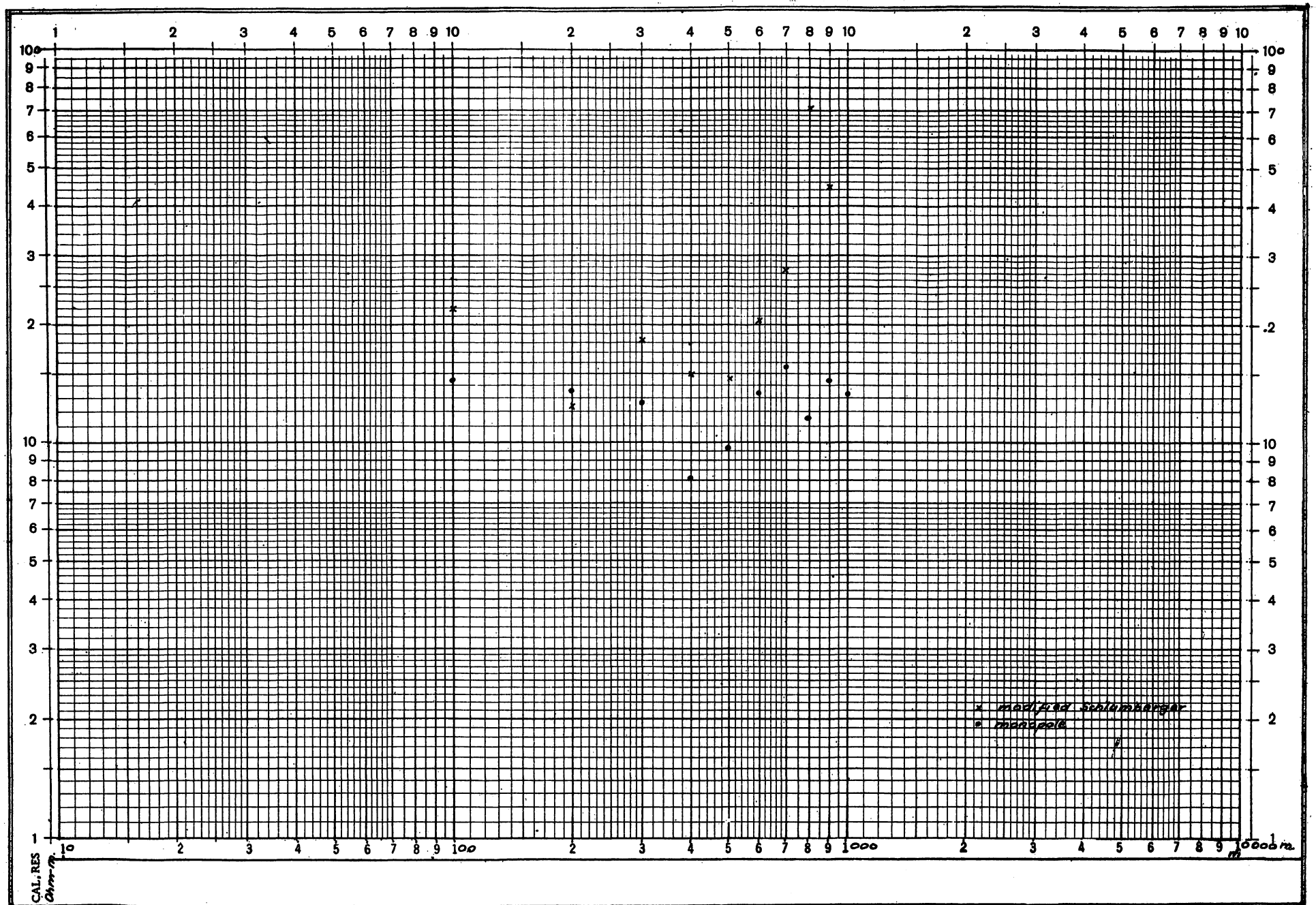
D.P. Nº. 53
FILE Nº. 001
JOB Nº. GT/149/74

LOCATION : EMEQ YIZREEL

COORDINATES : 170 100 - 224 050 169 950 - 225 550 ELEVATION :

(54088) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

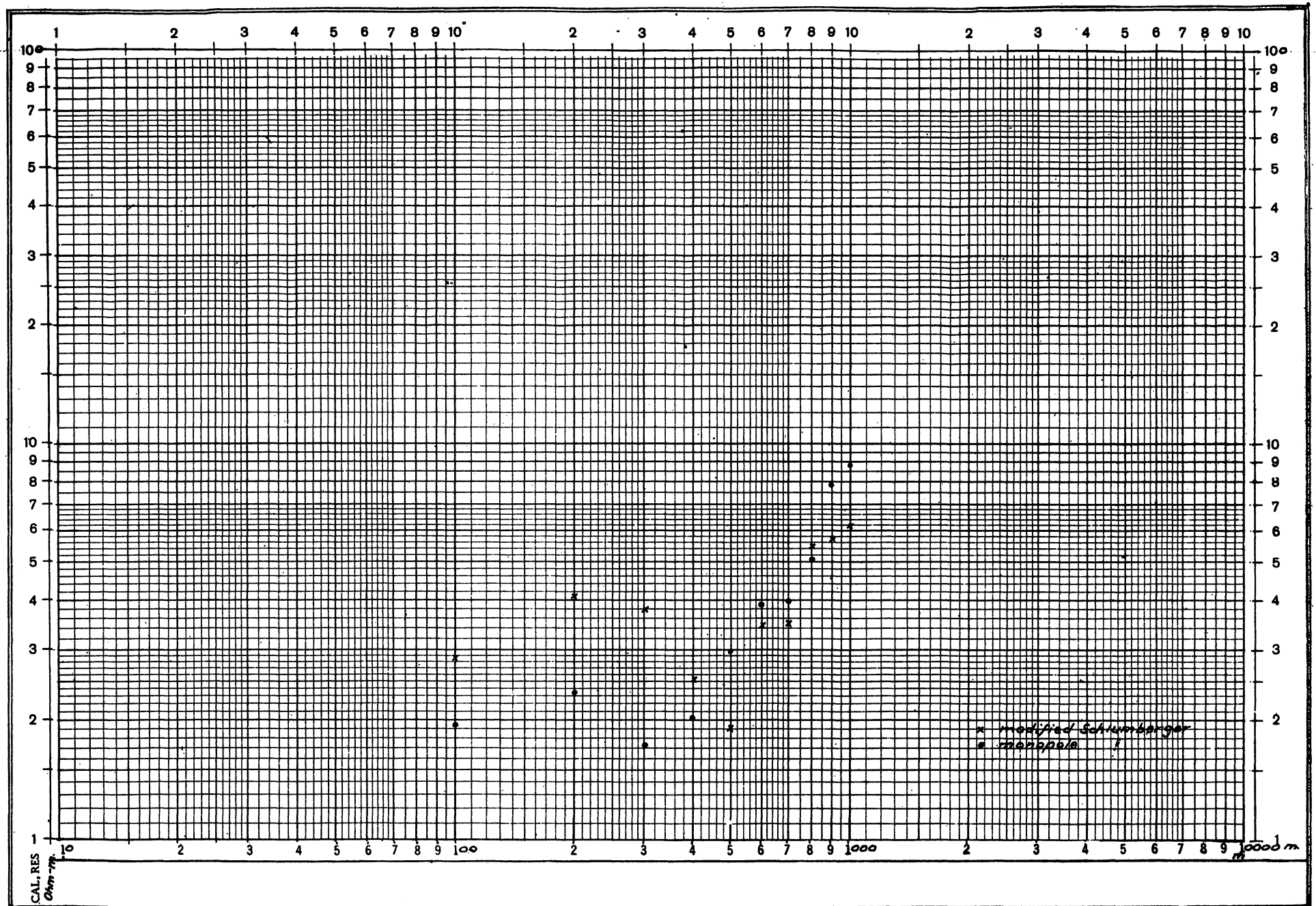
6,130



D.P. Nº 54
FILE Nº 001
JOB Nº QT//49/74

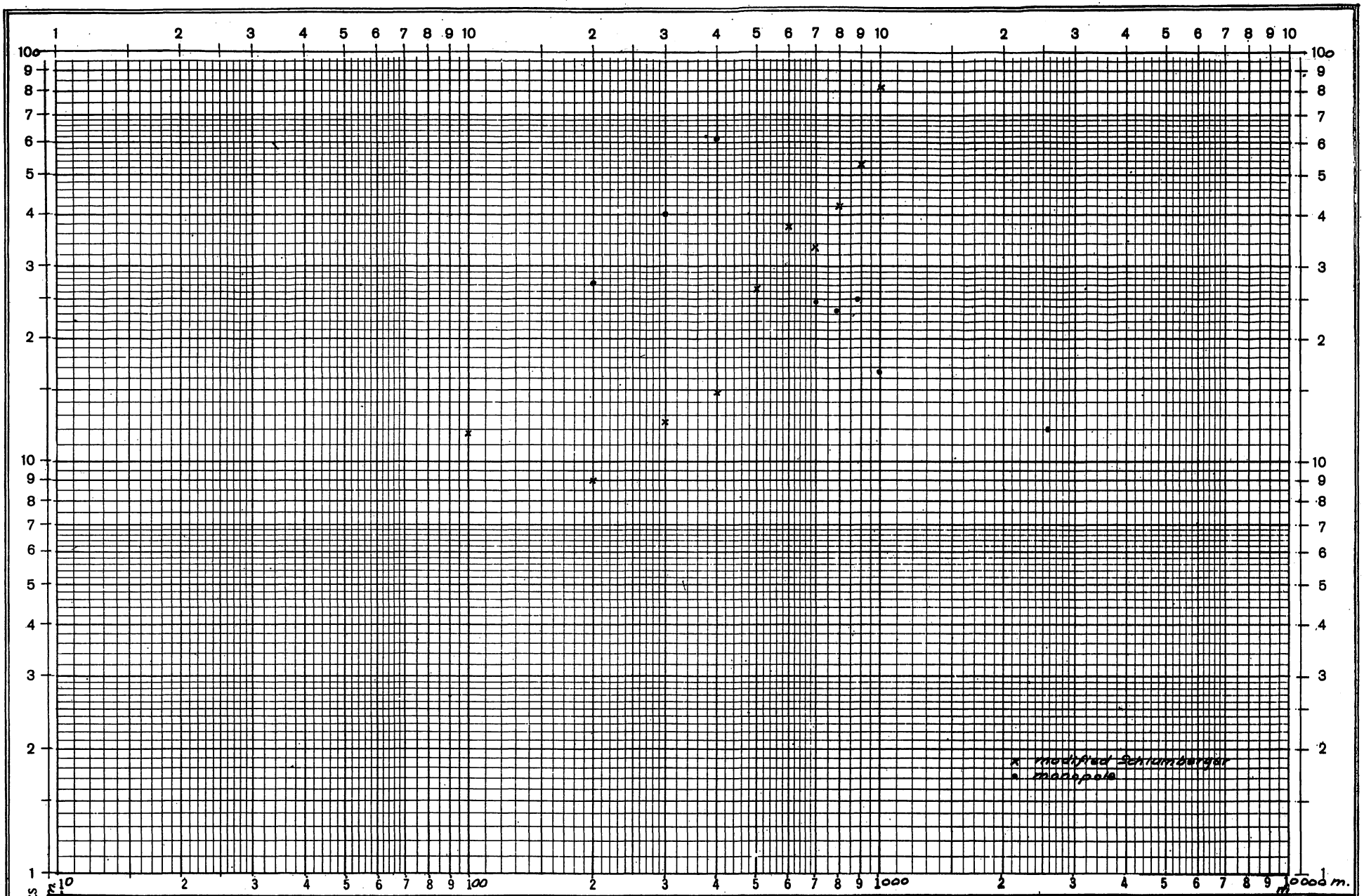
LOCATION : EMEQ YIZREEL
COORDINATES : 170100 - 224050 172300 - 225050 ELEVATION :
(54089) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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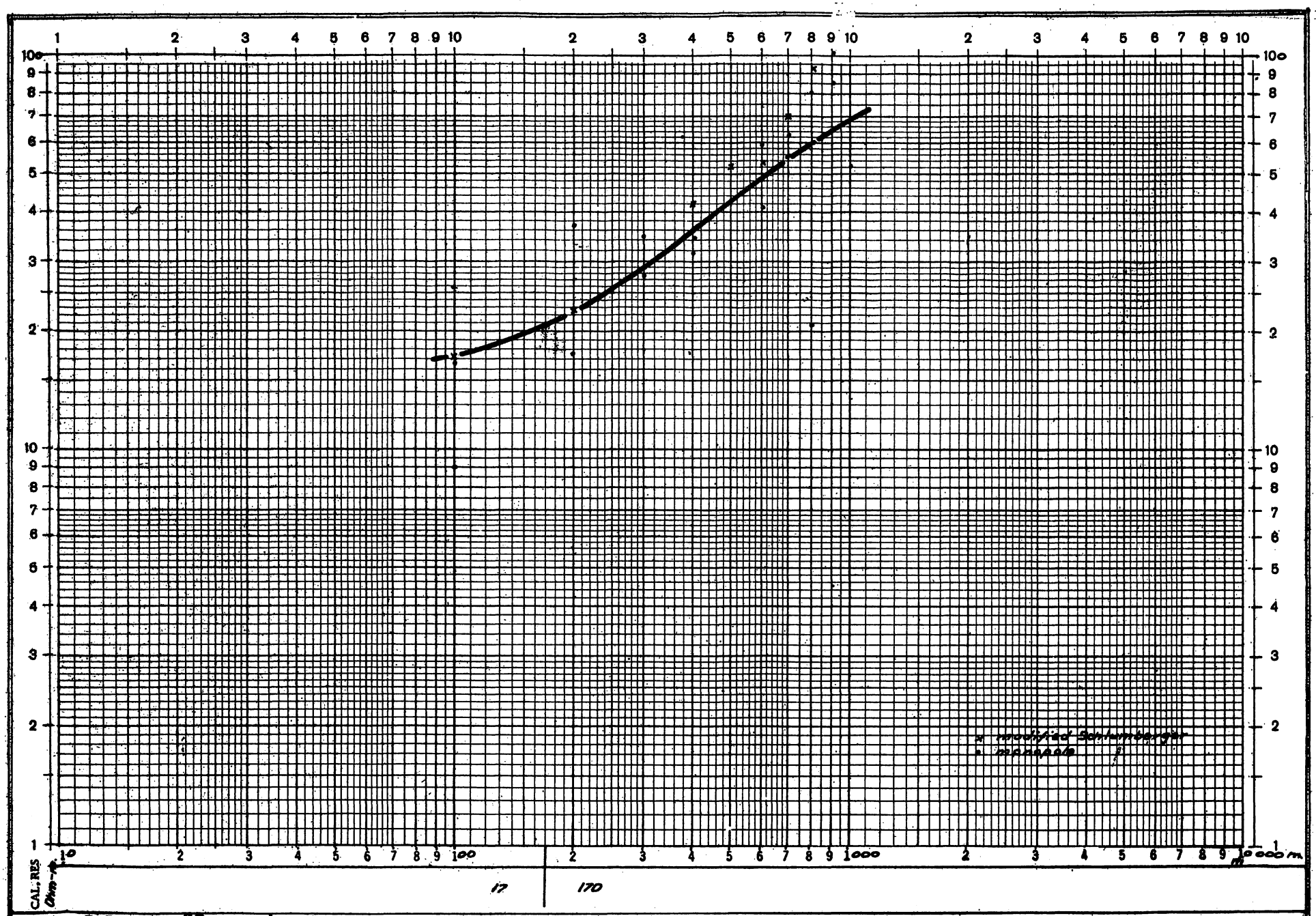
D.P. Nº . . 57 . .	LOCATION :	EMEQ YIZREEL
FILE Nº . 001 . .	COORDINATES :	184350 - 219900 184500 - 221800 ELEVATION :
JOB Nº GT/149/74	(54092)	THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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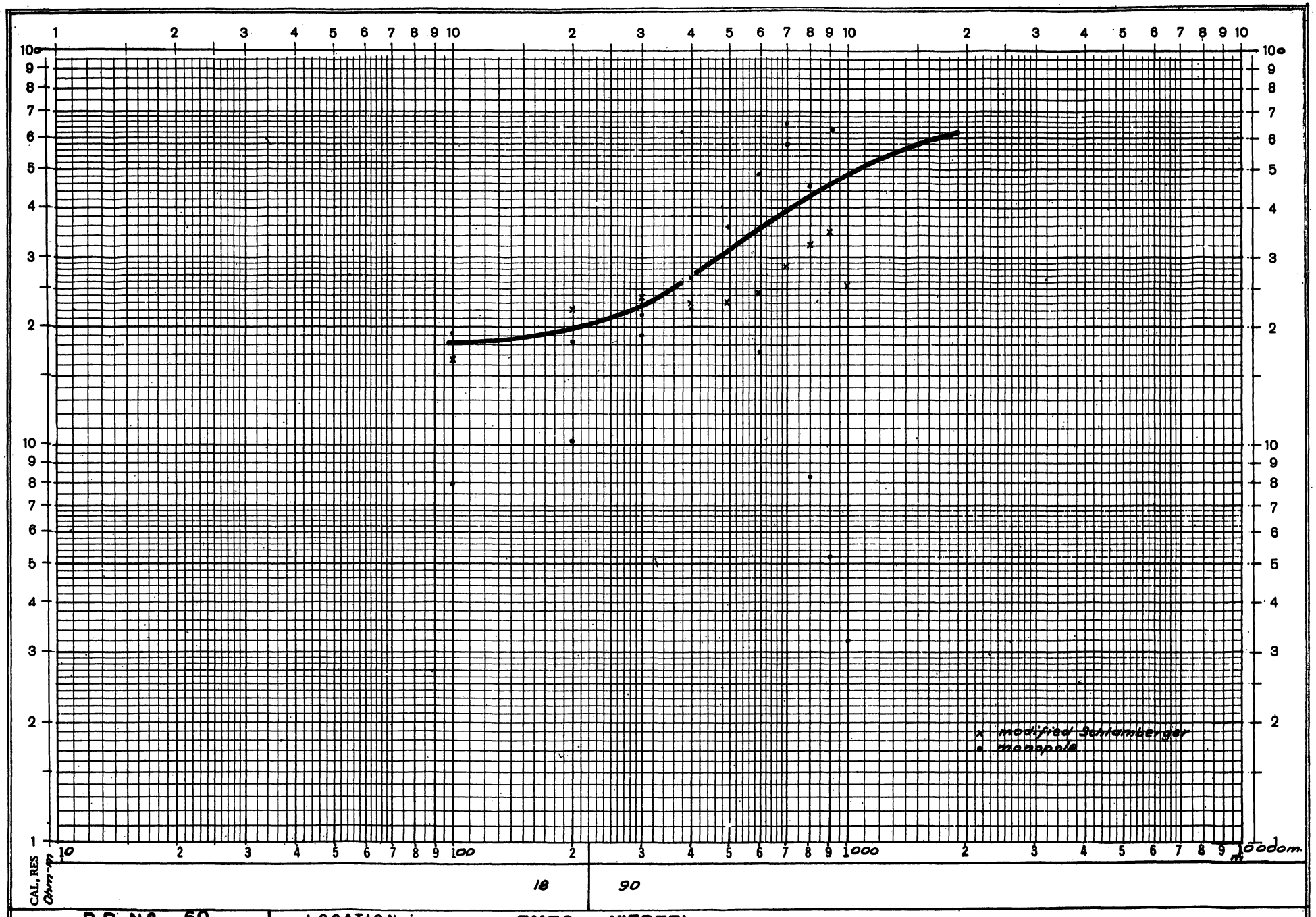


D.P. Nº 58	LOCATION : EMEQ YIZREEL	
FILE Nº 001	COORDINATES : 184350 - 219900	182960 - 219600 ELEVATION :
JOB Nº GT/149/74	(54093) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS	

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* modified Schlumberger
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D.P. Nº. 59.
FILE Nº. 001.
JOB Nº. 07/149/74
LOCATION : EMEQ YIZREEL
COORDINATES : 186700 - 215850 187300 - 217250 ELEVATION :
(540934) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

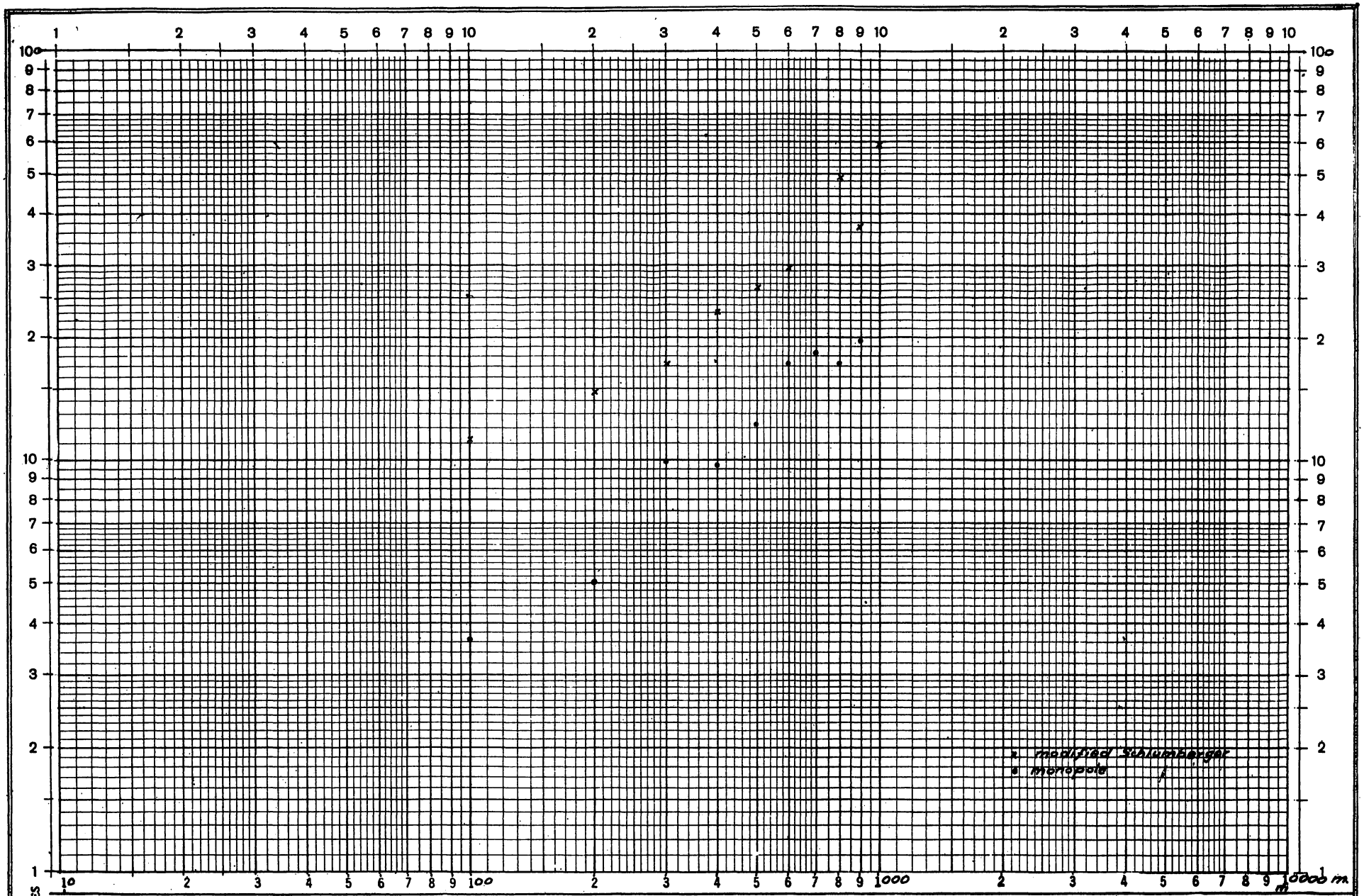
6.130



D.P. Nº. 60.
FILE Nº. 001.
JOB Nº. GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 186700 - 215850 185000 - 216980 ELEVATION :
(54095) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

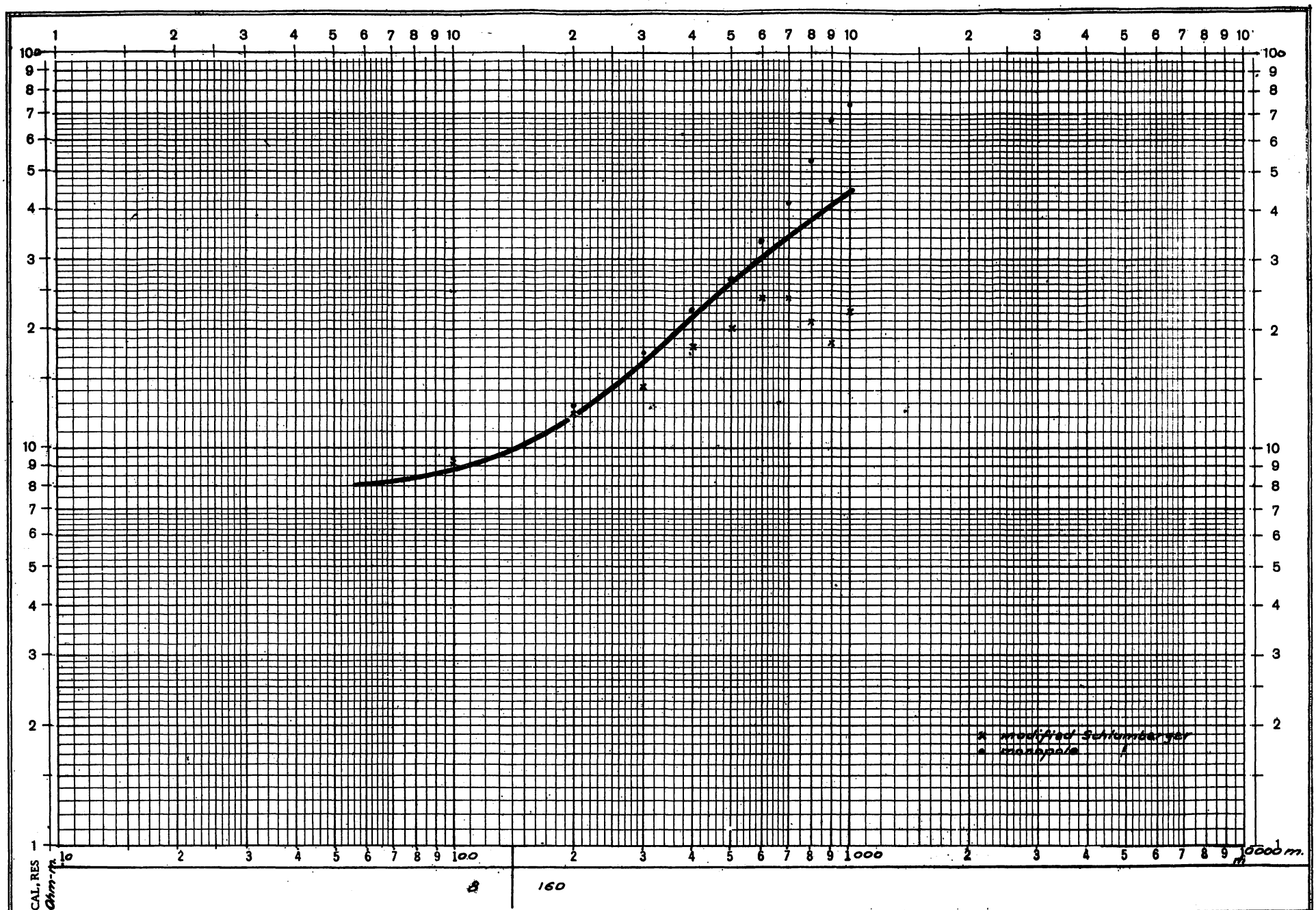
6,130



D.P. Nº . . 61 .
FILE Nº . 001 . .
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 191200 - 216400 180600 - 214800 ELEVATION :
(54096) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

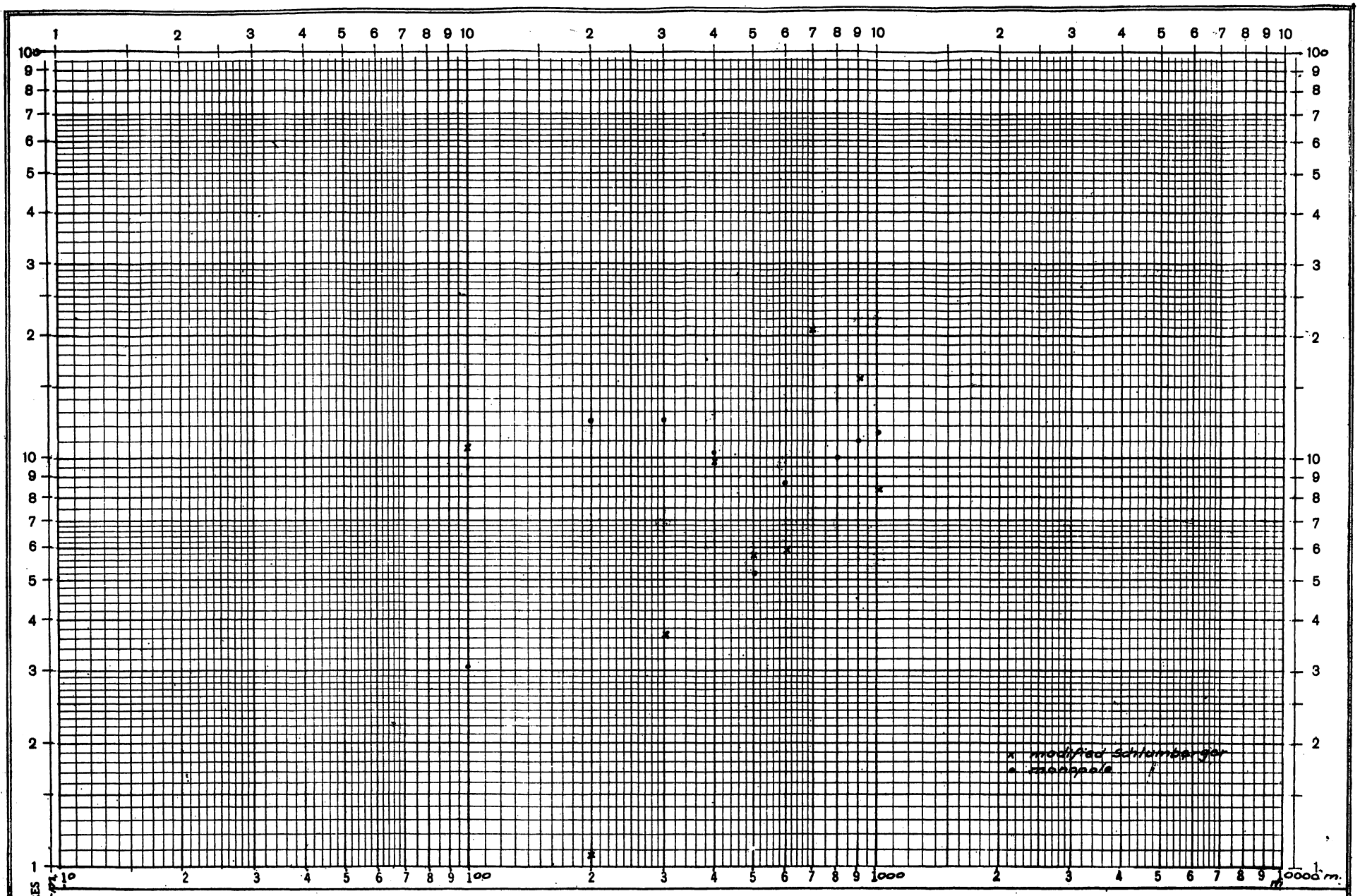
6,130



D.P. Nº. 62.
FILE Nº. 091.
JOB Nº. GT/149/74

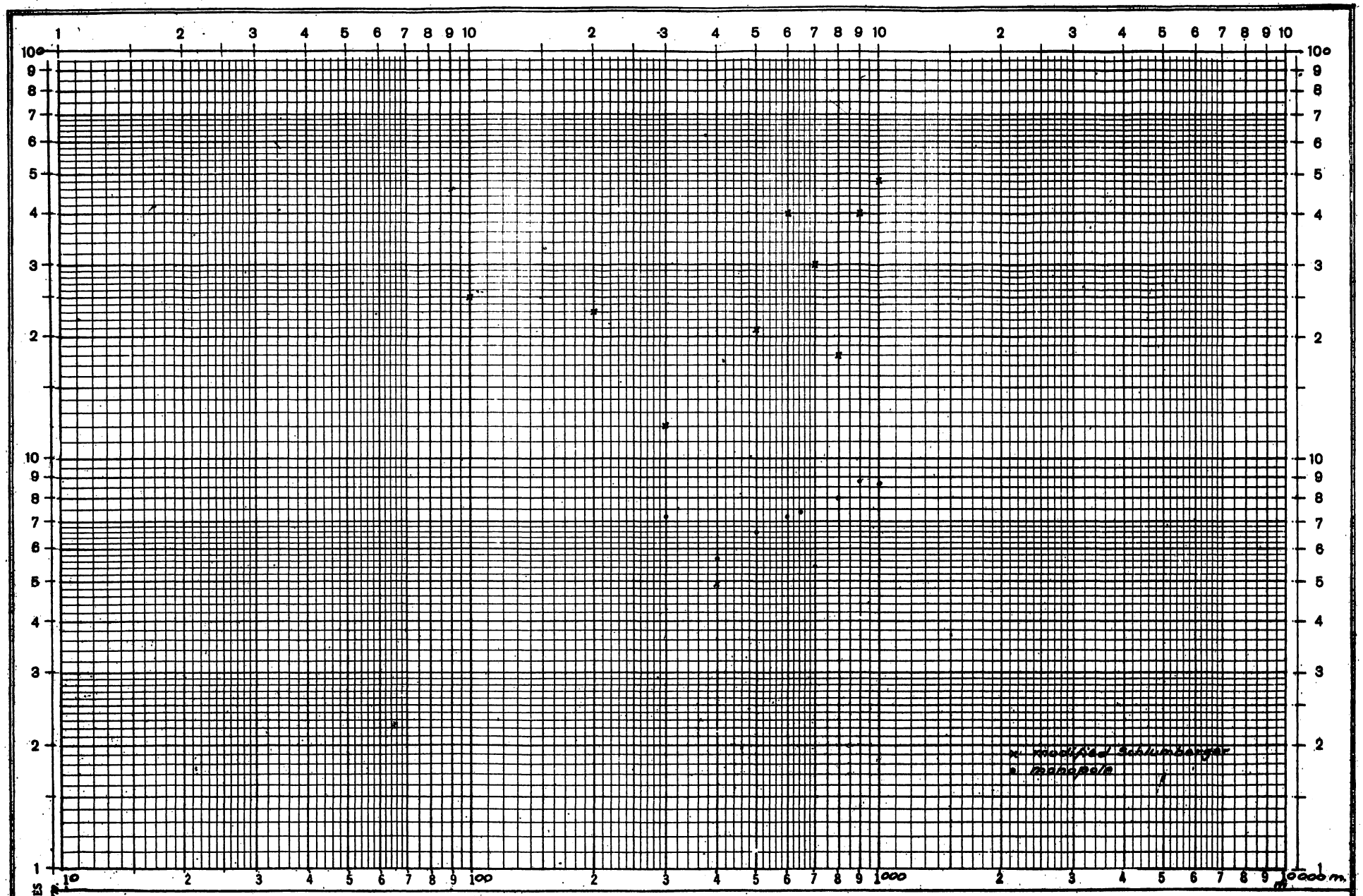
LOCATION : EMEQ YIZREEL
COORDINATES : 192900-215360 191200-216400 ELEVATION :
(54097) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

6.130



D.P. Nº. 63	LOCATION : EMEQ YIZREEL	
FILE Nº. 001	COORDINATES : 197000 - 217200	196200 - 215650 ELEVATION :
JOB Nº. GT/149/74	(54098) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS	

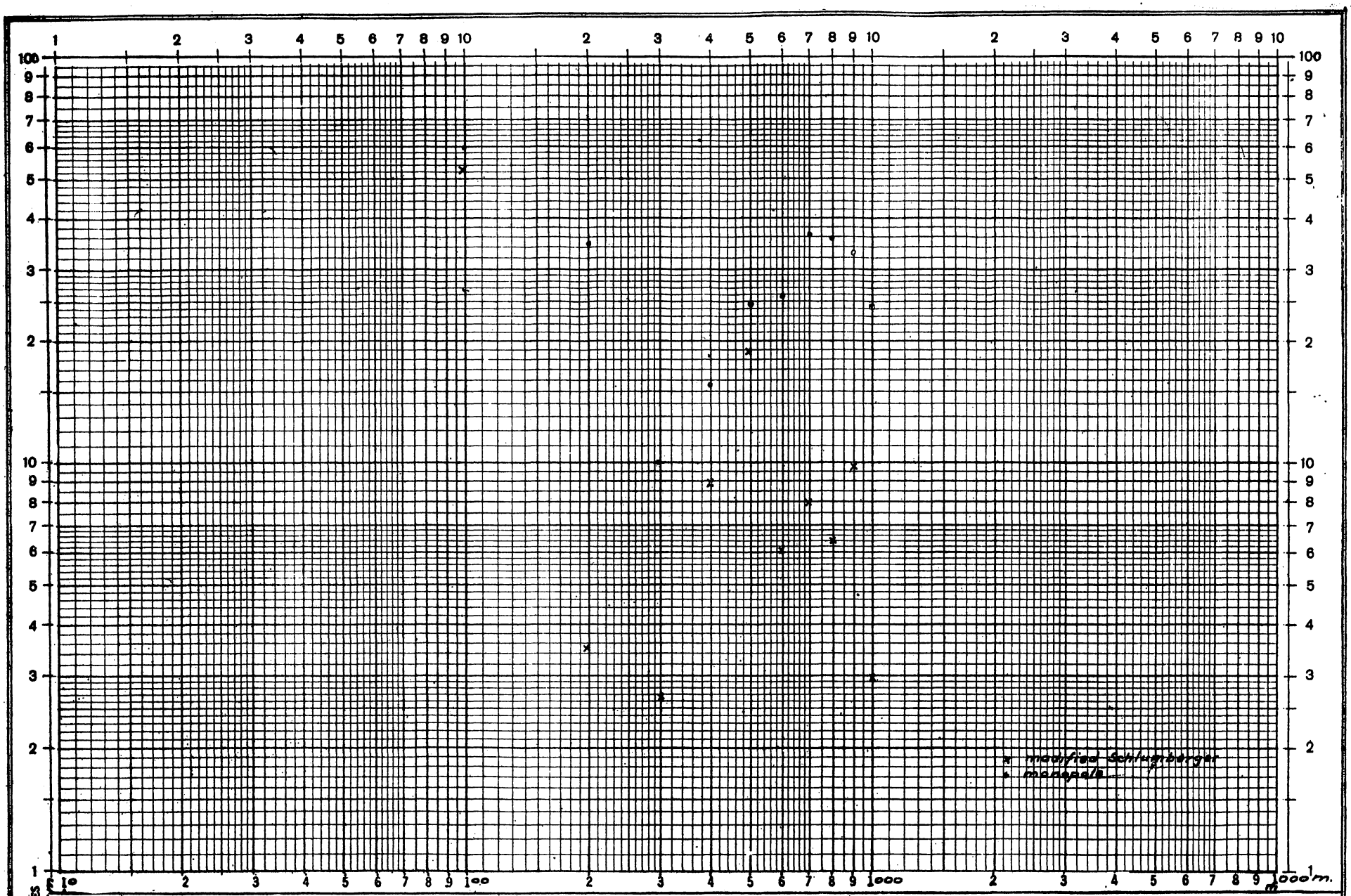
6,130



D.P. Nº 64
FILE Nº 001
JOB Nº 97/149/74

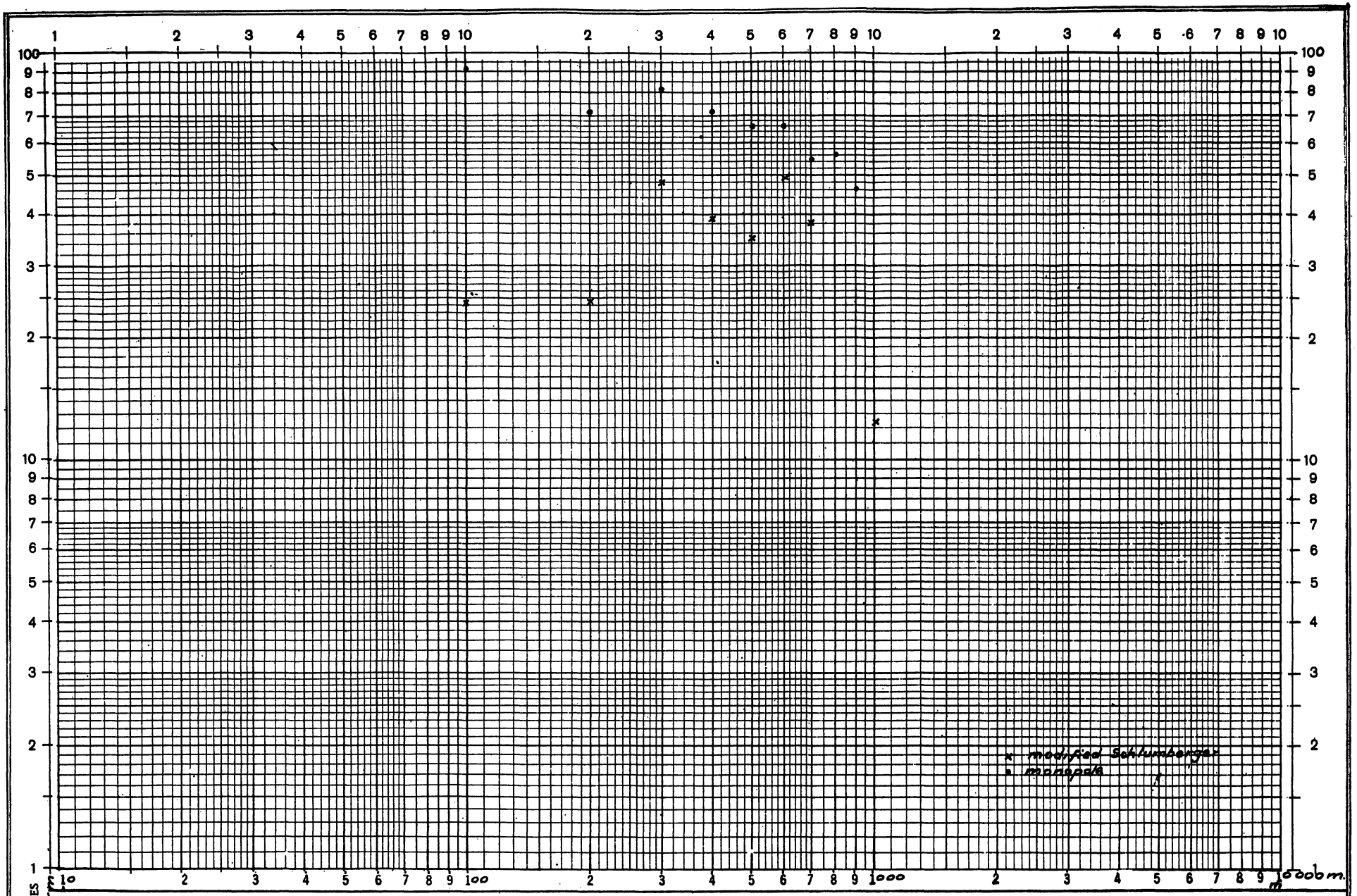
LOCATION : EMEQ YIZREEL
COORDINATES : 197000 - 217200 195550 - 218400 ELEVATION :
(54099) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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D.P. Nº. 65.	LOCATION :	EMEQ YIZREEL
FILE Nº 001	COORDINATES :	199900 - 214300 ELEVATION :
JOB Nº GT/149/74	(54100)	THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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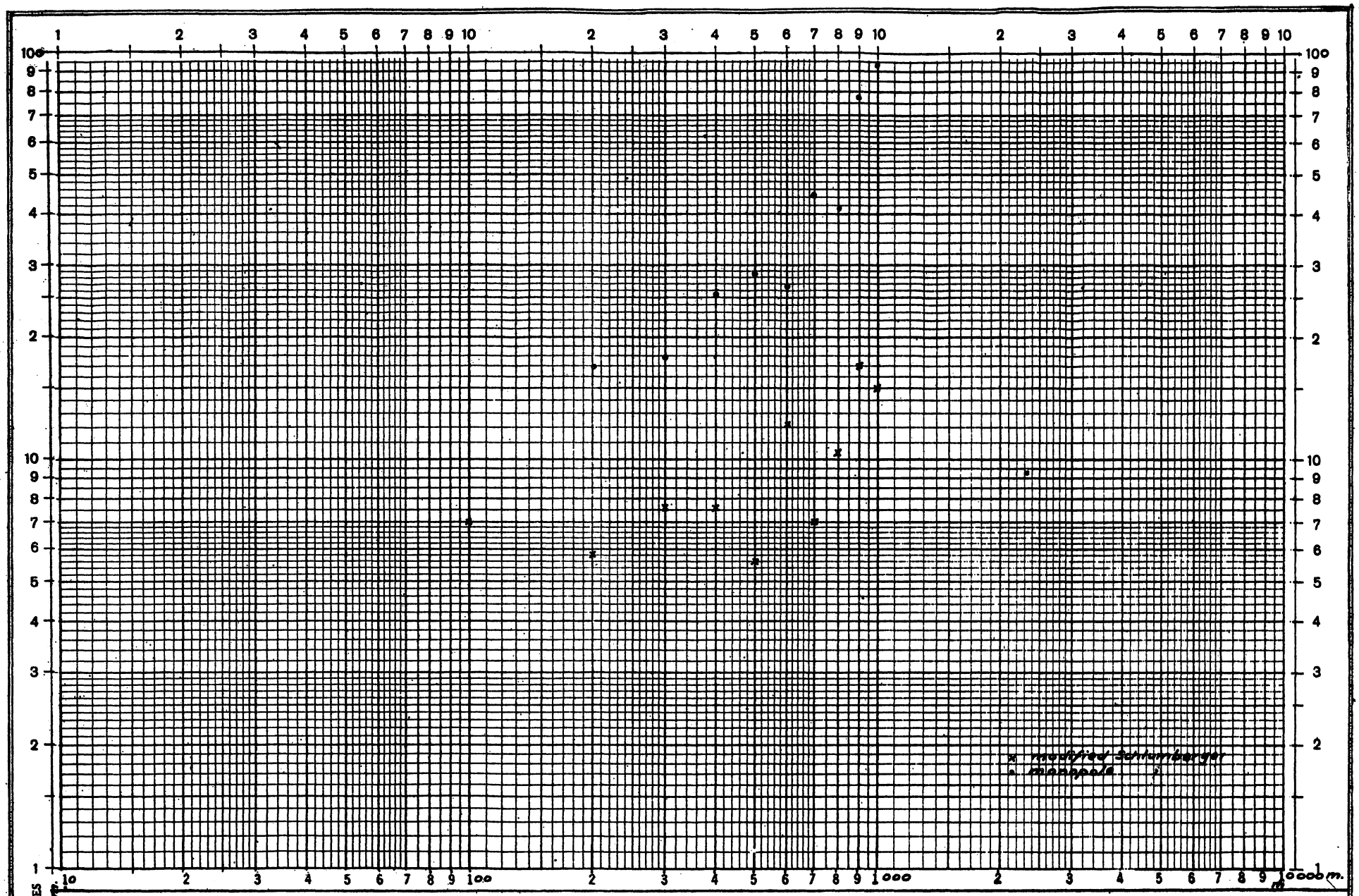
D.P. Nº 66
FILE Nº 001
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL

COORDINATES : 197400 - 213150 189000 - 212700 ELEVATION :

(54101) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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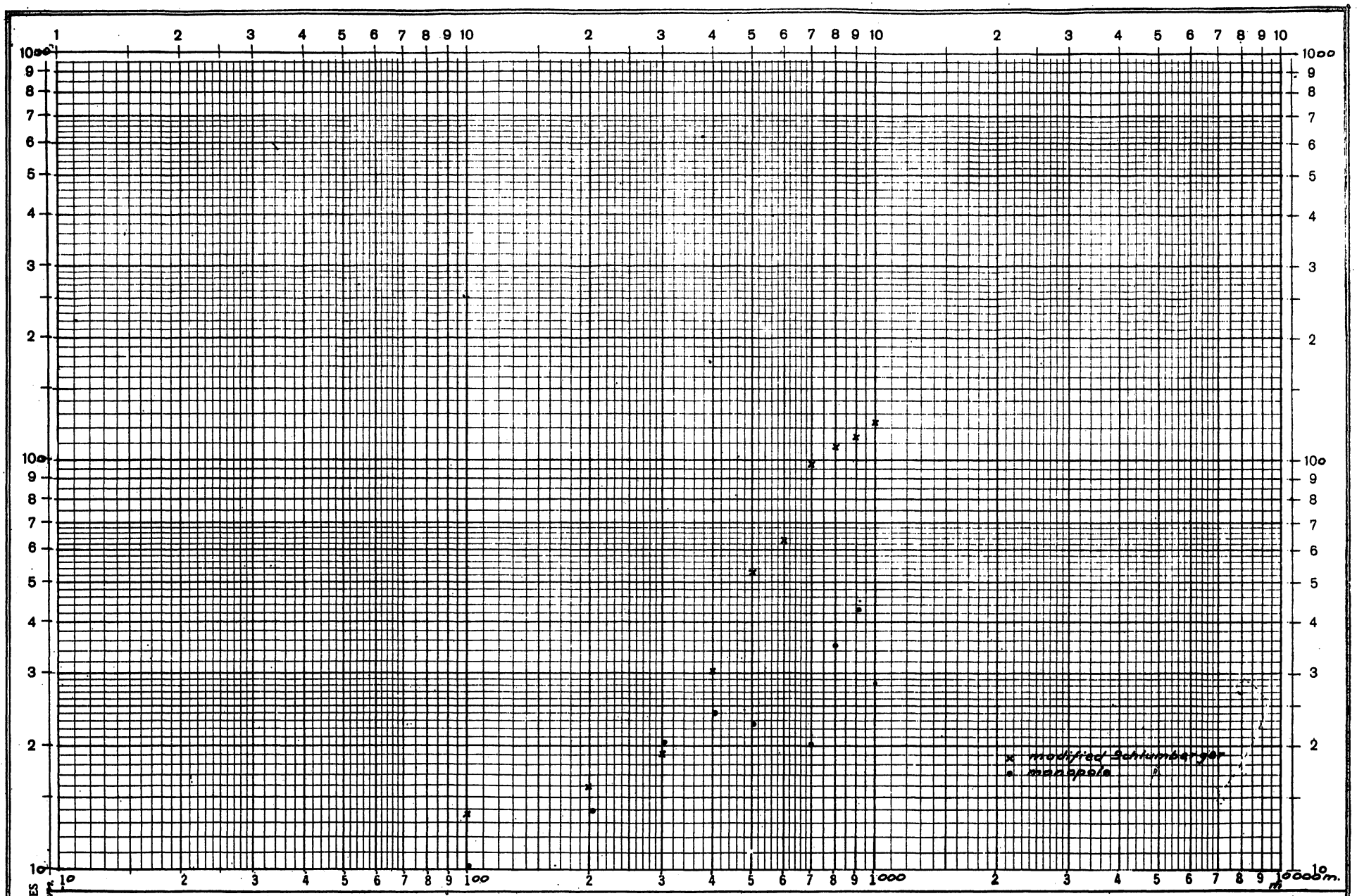
D.P. Nº. 71..
FILE Nº. 901..
JOB Nº. GT/149/74

LOCATION : EMEQ YIZREEL

COORDINATES : 163550 - 223200 163900 - 234900 ELEVATION :

(54105) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

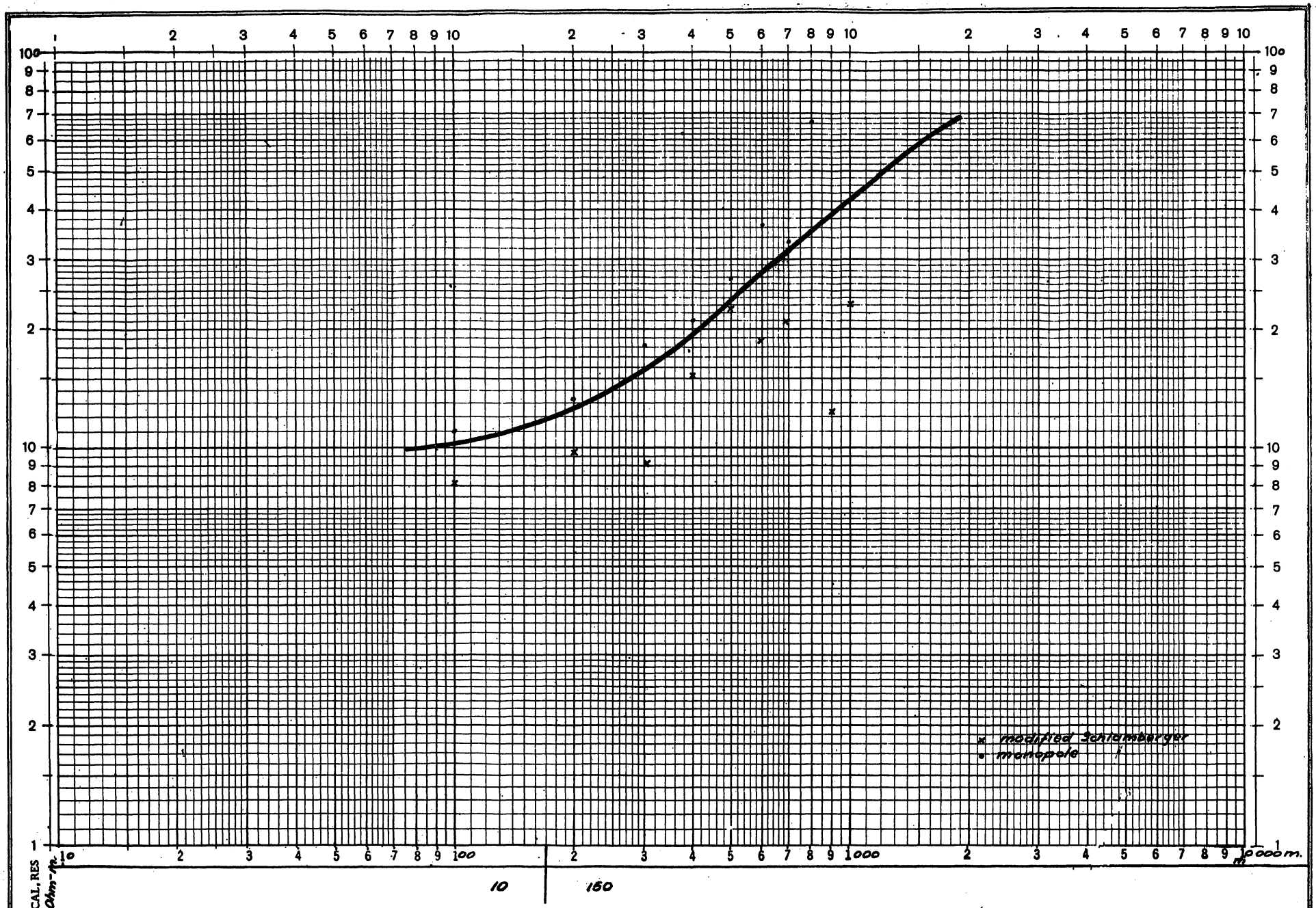
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D.P. Nº. 72..
FILE Nº. 001.
JOB Nº. GT/149/74

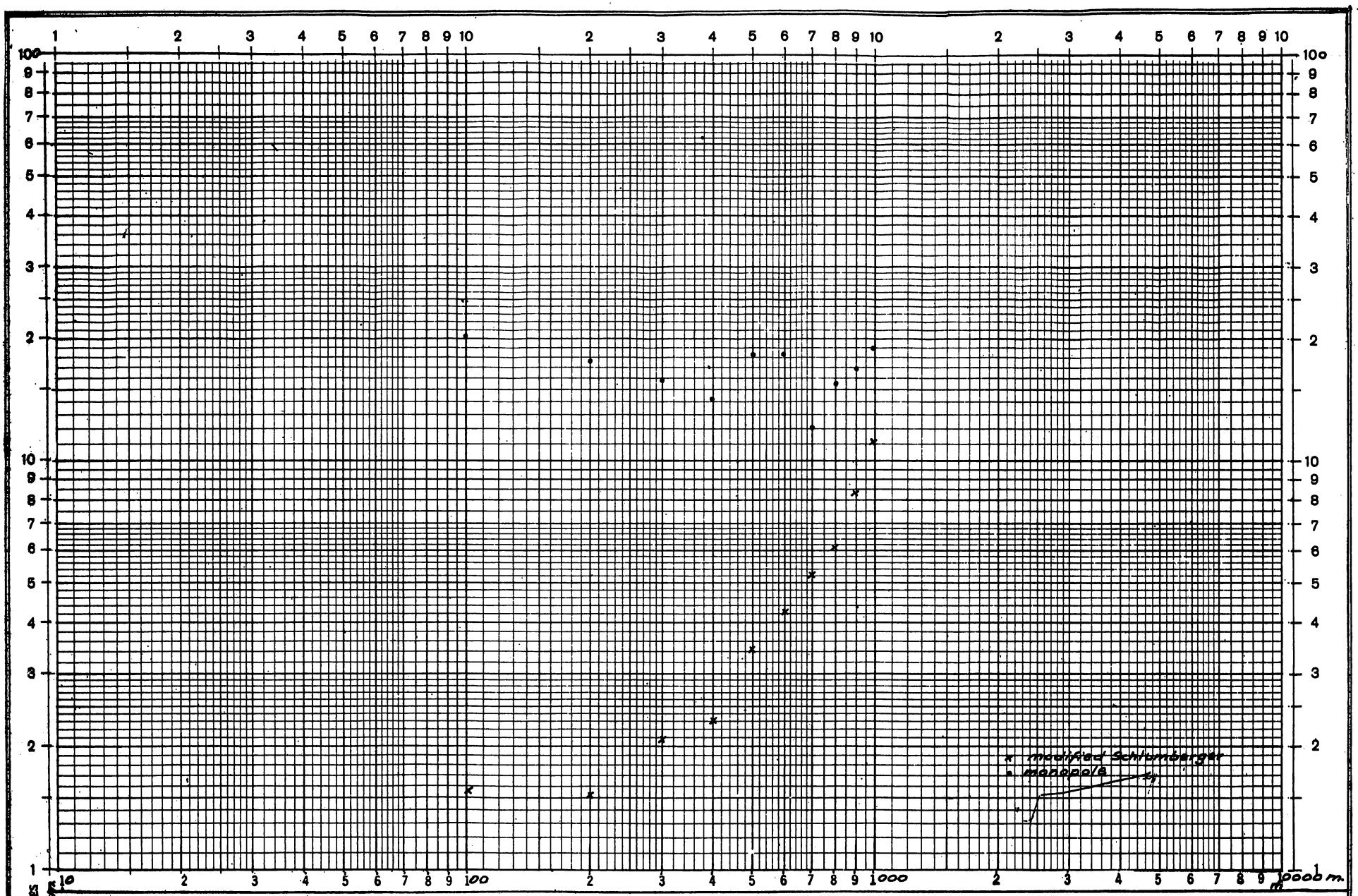
LOCATION : EMEQ YIZREEL
COORDINATES : 163 900 - 234 900 ELEVATION :
(54107) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

6,130



D.P. Nº 73	LOCATION : EMEQ YIZREEL	
FILE Nº 001	COORDINATES : 177650 - 229300	177460 - 231820 ELEVATION :
JOB Nº GT/149/74	(54108) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS	

6.130



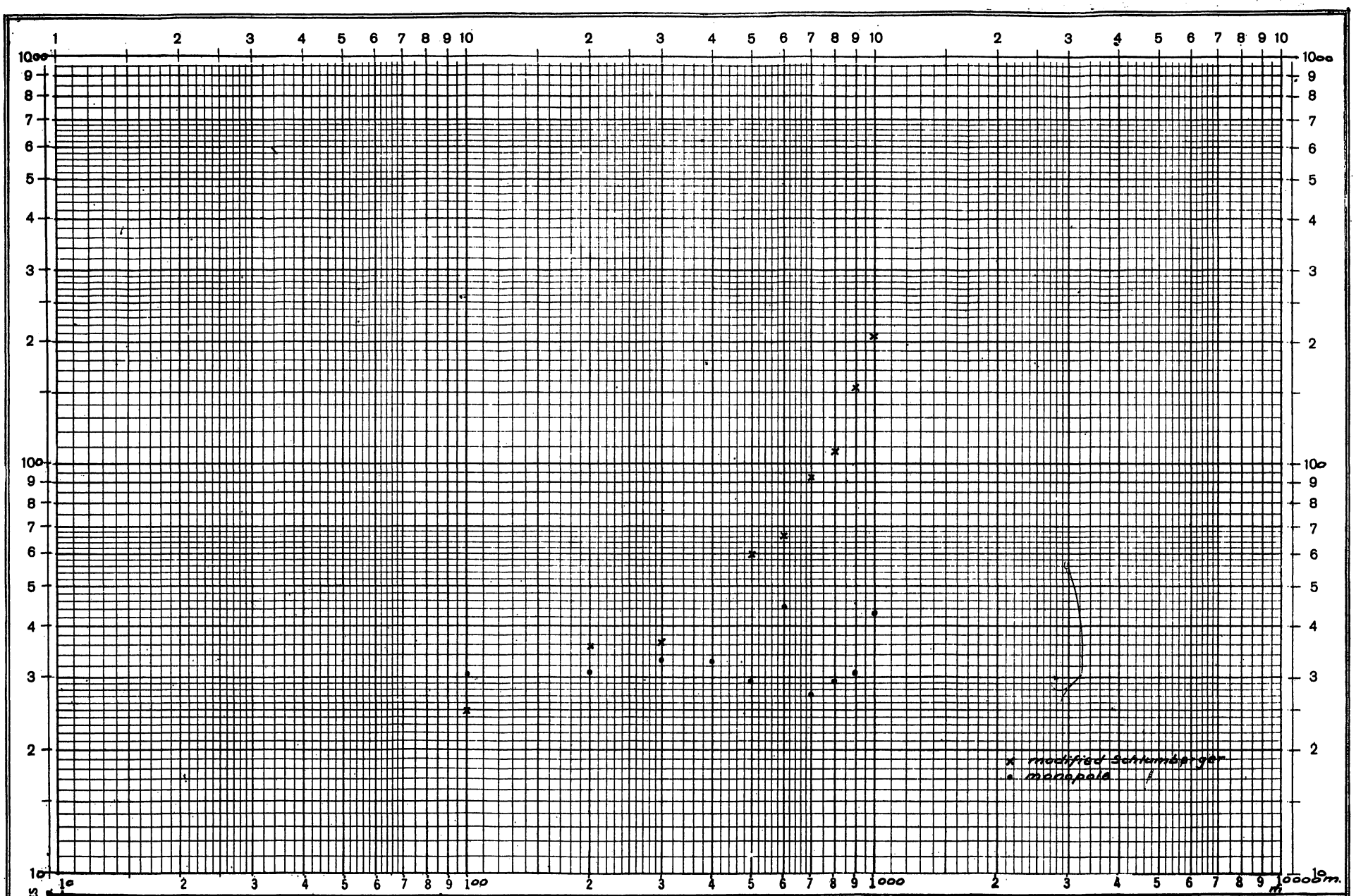
D.P. Nº. 74..
FILE Nº. 001..
JOB Nº 67/149/74

LOCATION : EMEQ YIZREEL

COORDINATES : 175700 - 229520 177650 - 229300 ELEVATION :

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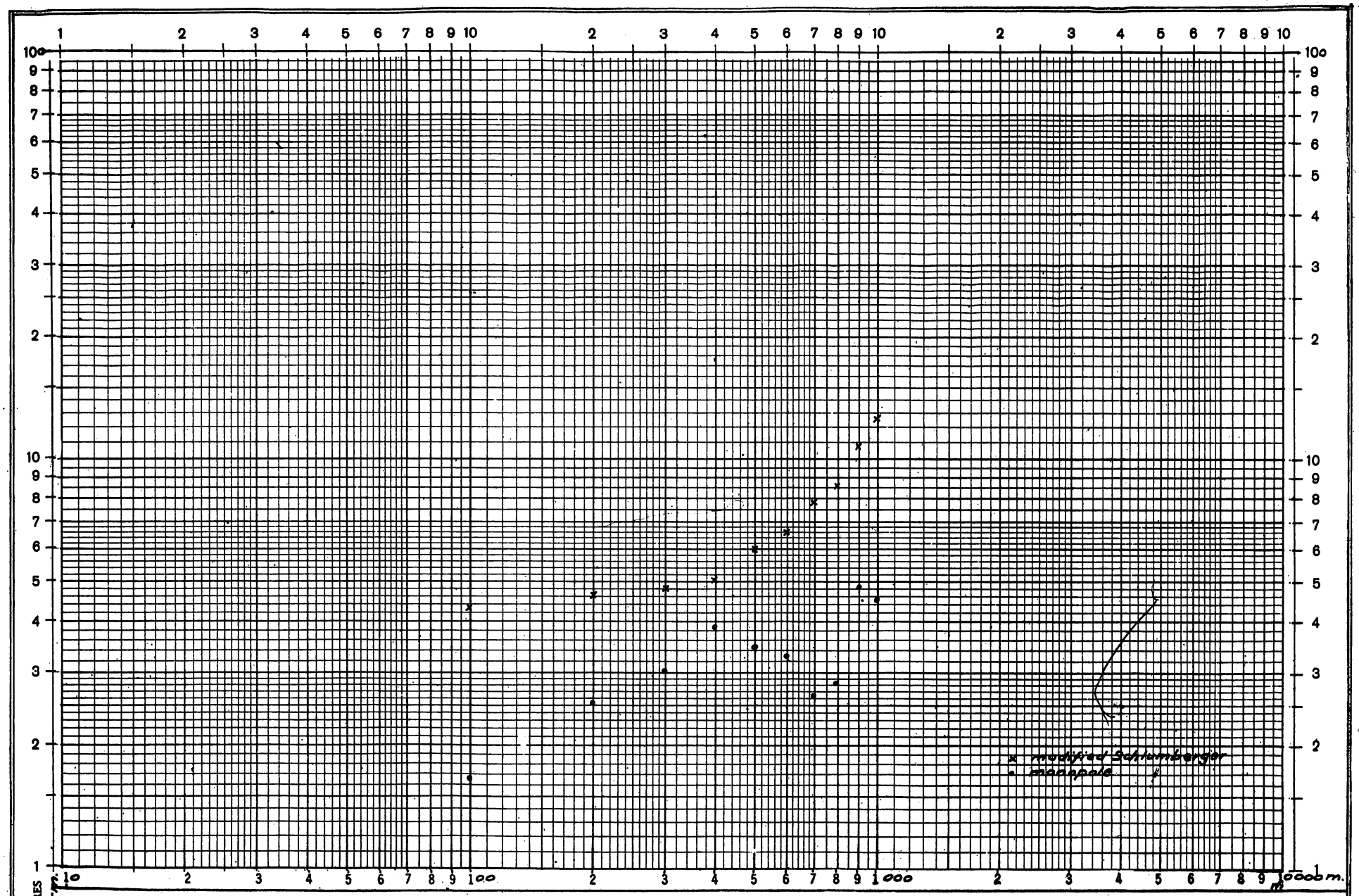
6.130.



D.P. Nº 75
FILE Nº 001
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 186200 - 231150 ELEVATION :
(54110) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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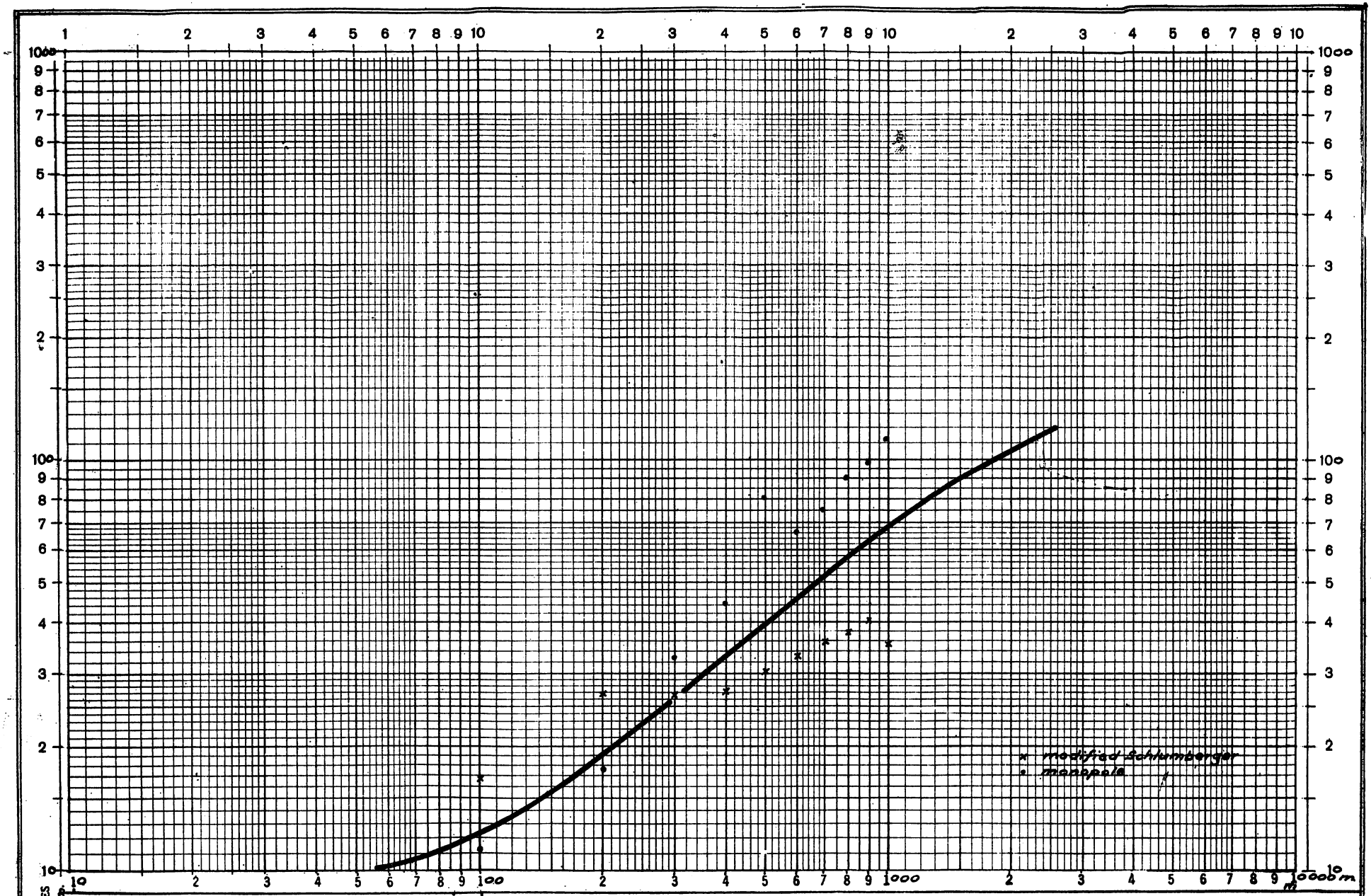
D.P. Nº. 76..
FILE Nº. 001..
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL

COORDINATES : 185400 - 228650 187200 - 229850 ELEVATION :

(54111) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

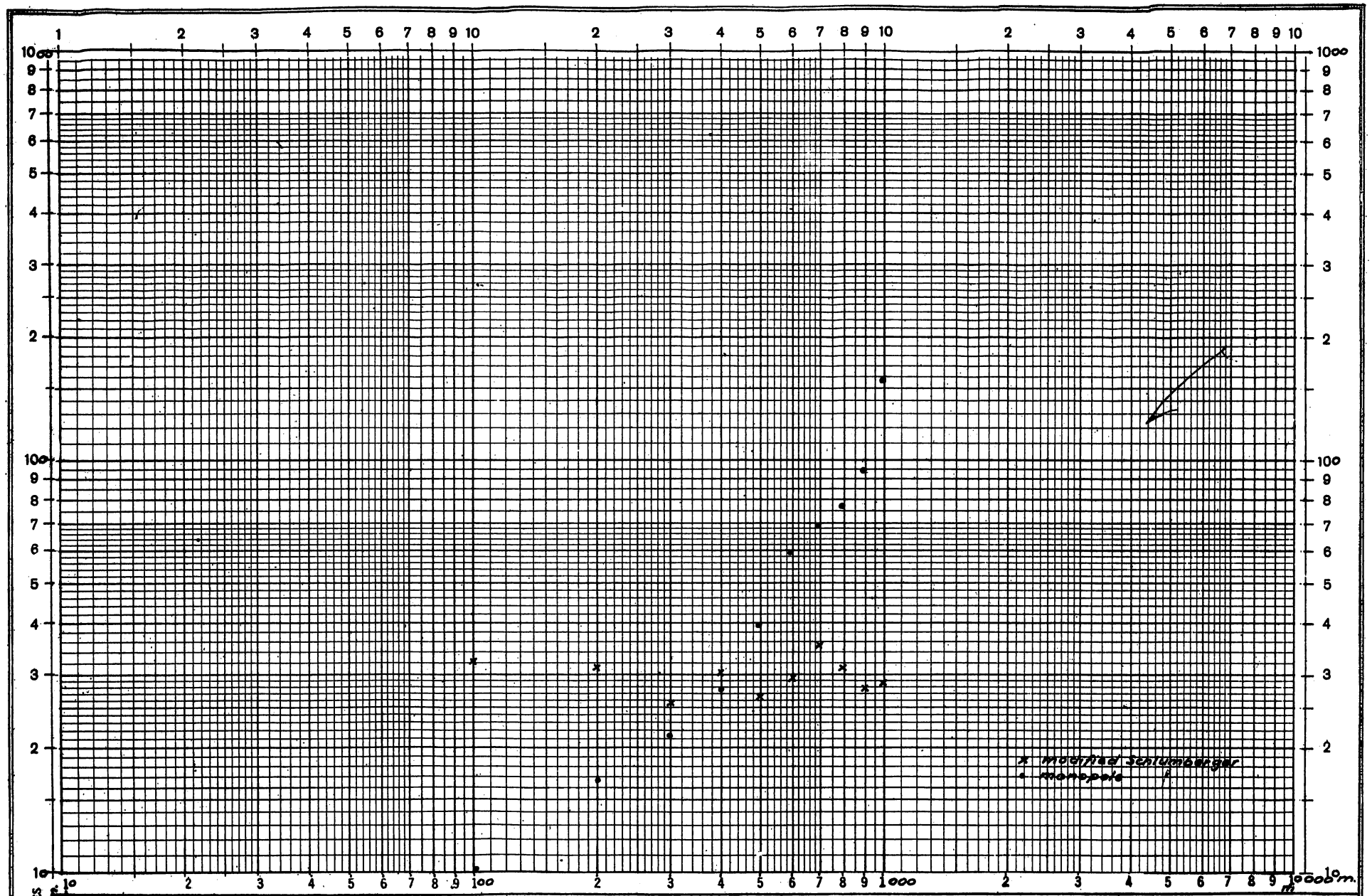
6.130



D.P. Nº 77
FILE Nº 001
JOB Nº GT/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 187800 - 220600 187450 - 222450 ELEVATION :
(54112) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

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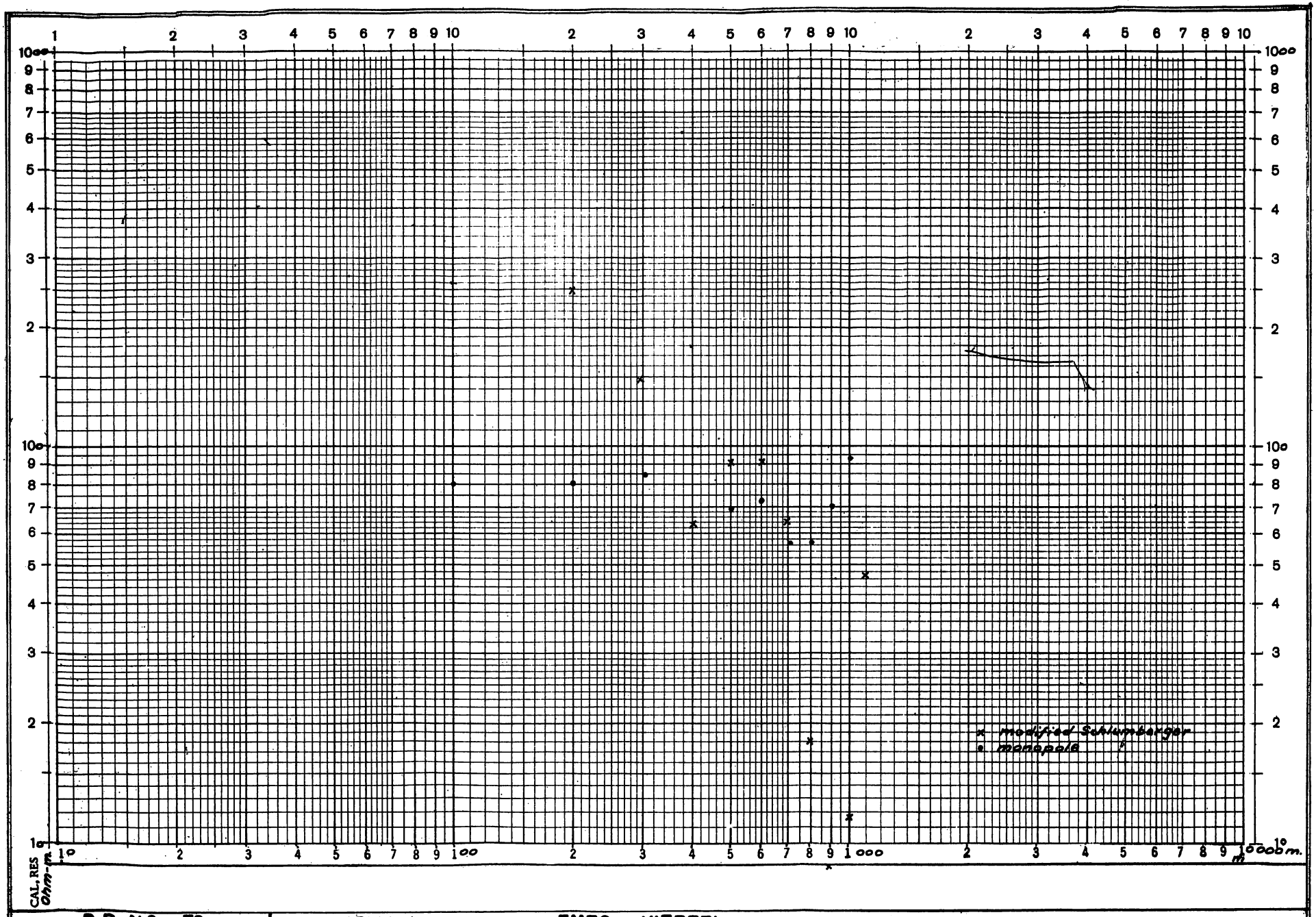


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D.P. Nº . 78 .
FILE Nº . 001 . .
JOB Nº . 67/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 187 800 - 220 500 189 650 - 221 450 ELEVATION :
(54113) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

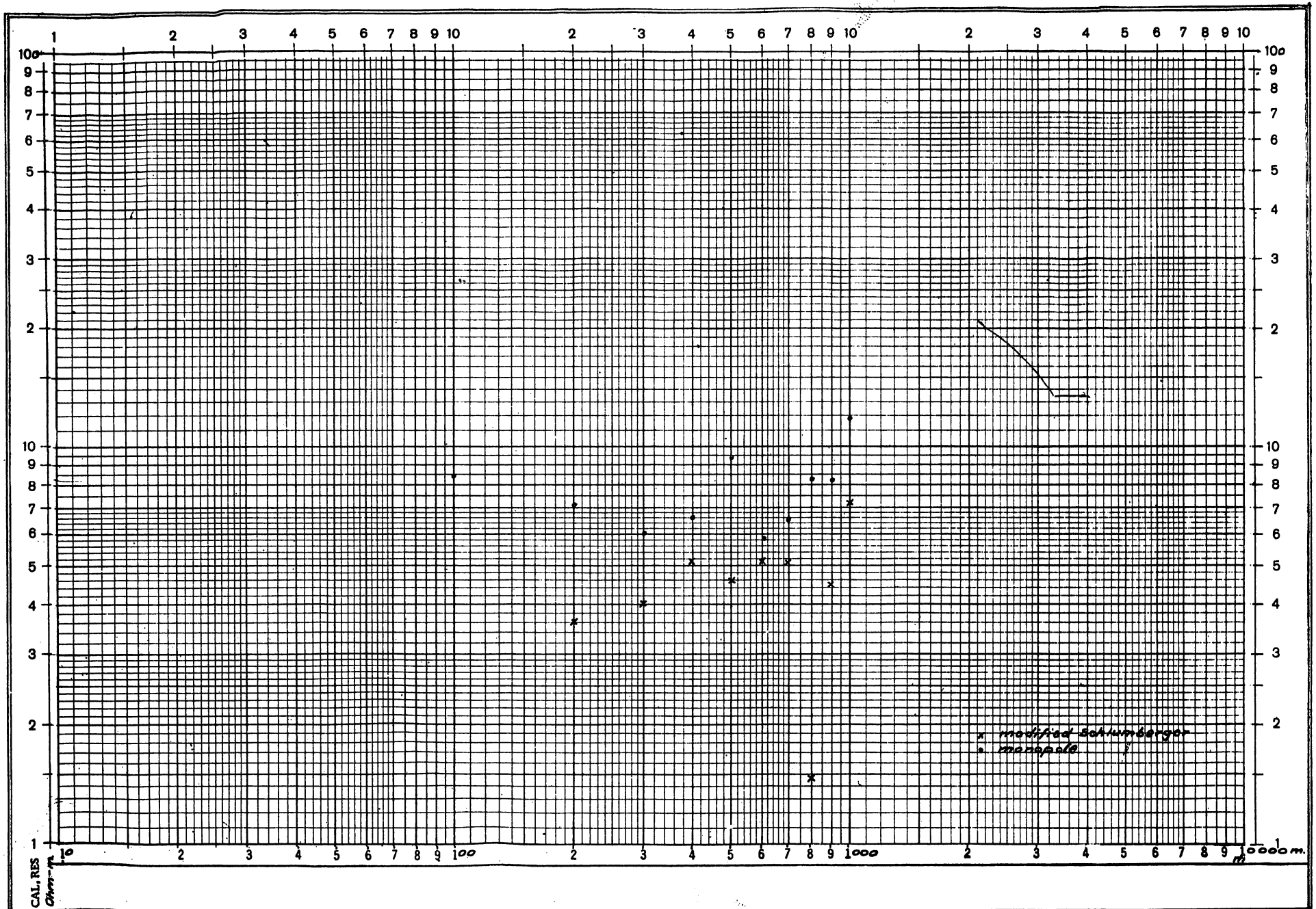
6.130



D.P. Nº . 79 .
FILE Nº . 001 .
JOB Nº 67/149/74

LOCATION : EMEQ YIZREEL
COORDINATES : 195500 - 222700 ELEVATION :
(54114) THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

6.130



D.P. Nº 80	LOCATION : EMEQ YIZREEL	
FILE Nº 001	COORDINATES : 193 500 - 223 200	ELEVATION :
JOB Nº 9T/149/74 (54115)	THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS	

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